



Quantum Café

A gathering place for science, ideas, and the curious

May 31st 2026 | MIT | The Engine



BOS TECH WEEK



we're
~~hosting~~
HACKING



MIT HOUSE



Weights
& Biases



CoreWeave

AI!

Where?

- Quantum Fundamentals
- Hardware
- Community



The background image shows a complex quantum computing hardware setup. On the left, there's a large, cylindrical metallic structure, possibly a cryostat or a part of a quantum processor. To its right, a red rectangular box is labeled "sees" and "0.0112". Further right, a vertical assembly of metal rods and components is visible. In the foreground, a square-shaped device with a white frame and a copper-colored outer ring is mounted. Inside the white frame, a small, colorful, rectangular object is visible. To the right of this device, there's a black rectangular component with a circular opening. The entire setup is connected by various wires and cables. The QuEra logo is overlaid on the left side of the image.

QuEra



Neutral Atom Quantum Computers - Concept of Operation | QuEra



QuEra Computing
2.09K subscribers



246



Share



Ask



Save



Download



Qbit

(Quantum bit)

Rubidium Atom



Qbit

$|0\rangle$ = ground state



(lower energy, “unexcited”)

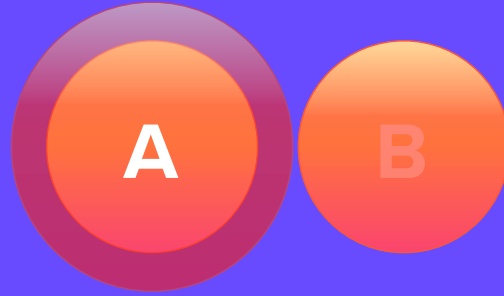
Qbit

$|1\rangle$ = excited state
(higher energy)

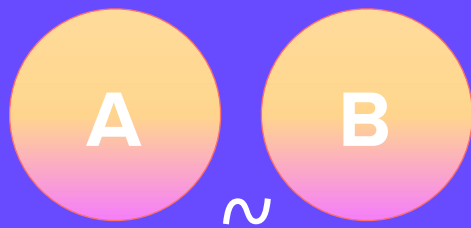


Rydberg
atoms

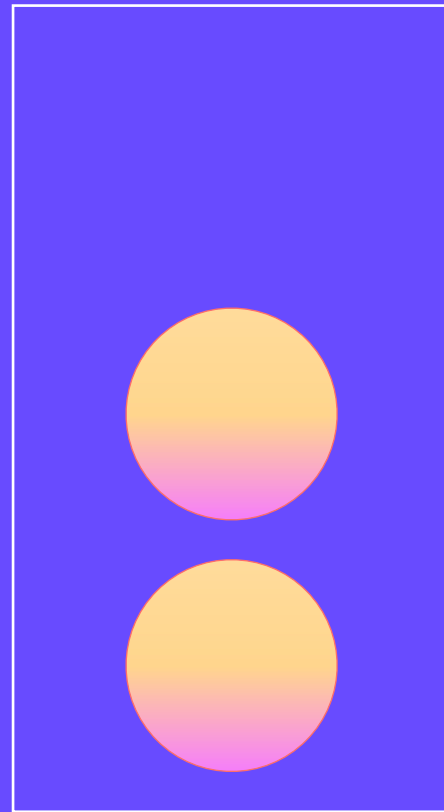




- Atoms A and B sit close together in their optical tweezers
- Laser excites atom A to the Rydberg state
- Atom A's Rydberg state shifts atom B's energy levels
- The same laser can no longer excite atom B
- B's evolution depended on A's state
- Their quantum states have become correlated
- They can no longer be described independently
- That's entanglement!



Preserved

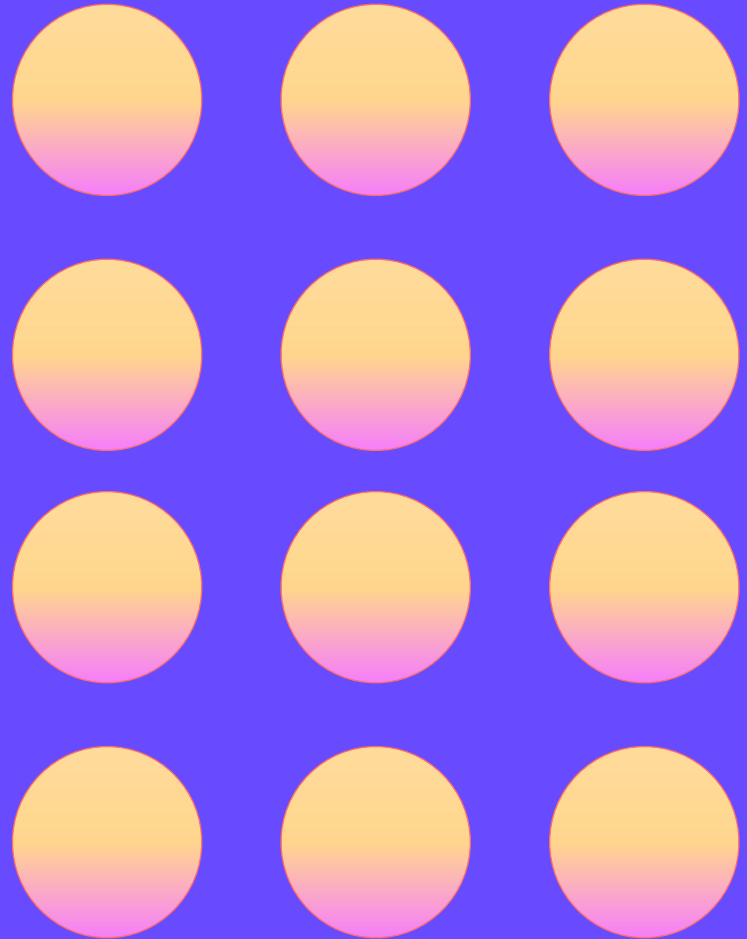


Read out

QPU

(Quantum

Processing Unit)



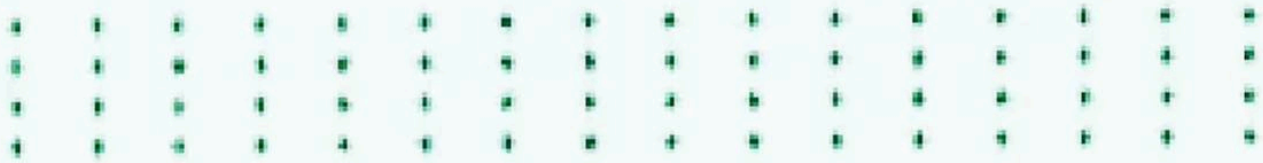


T: μK

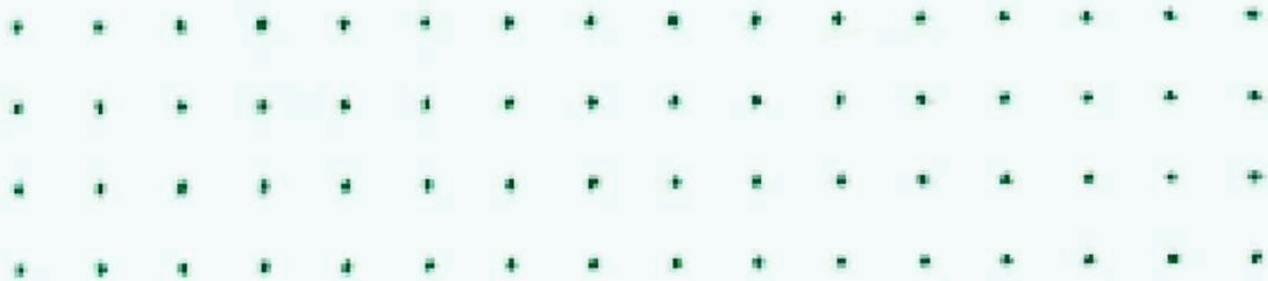
**MolSim: anything below..
breaks the simulation!**

Atom Shuttling

Storage

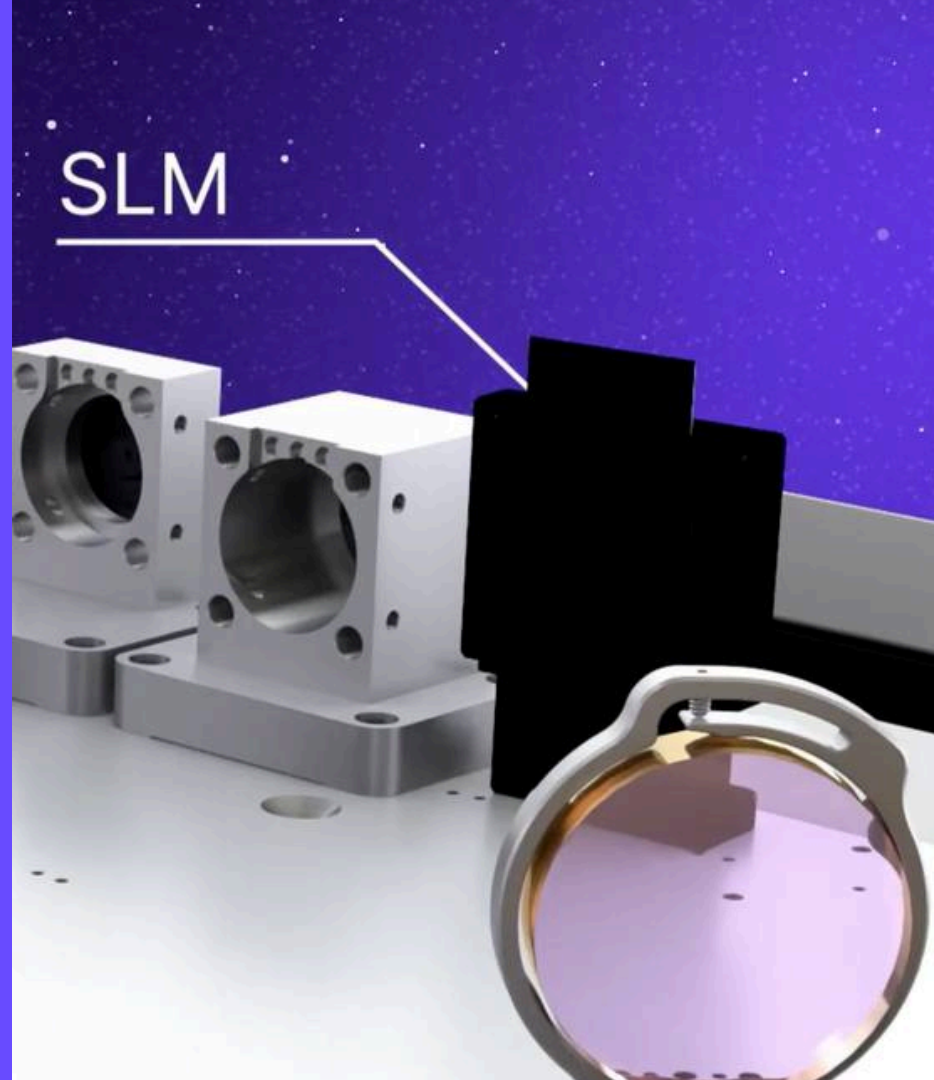


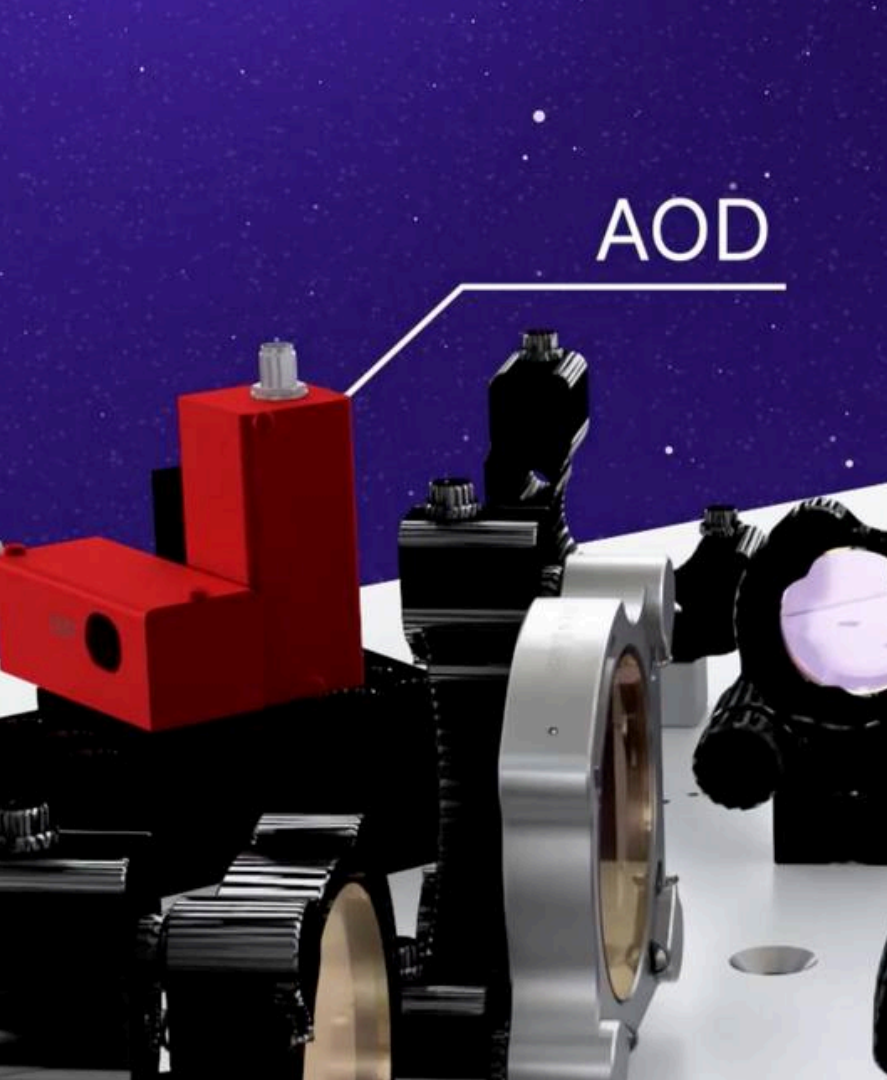
Entanglement



SLM

(Spatial Light
Modulator)

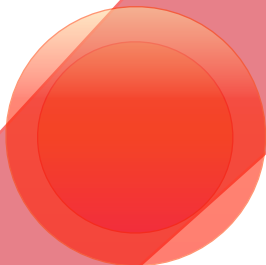
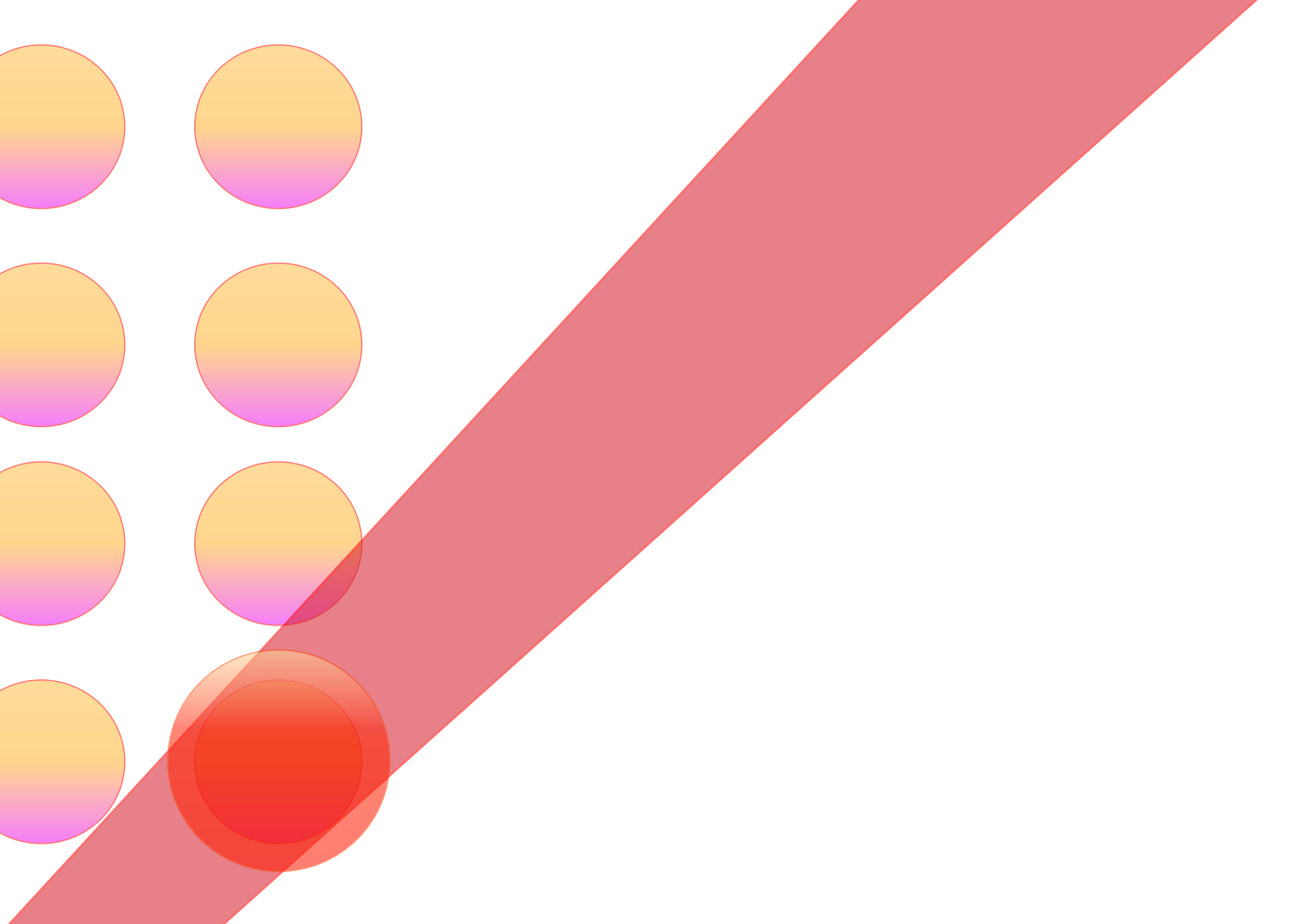
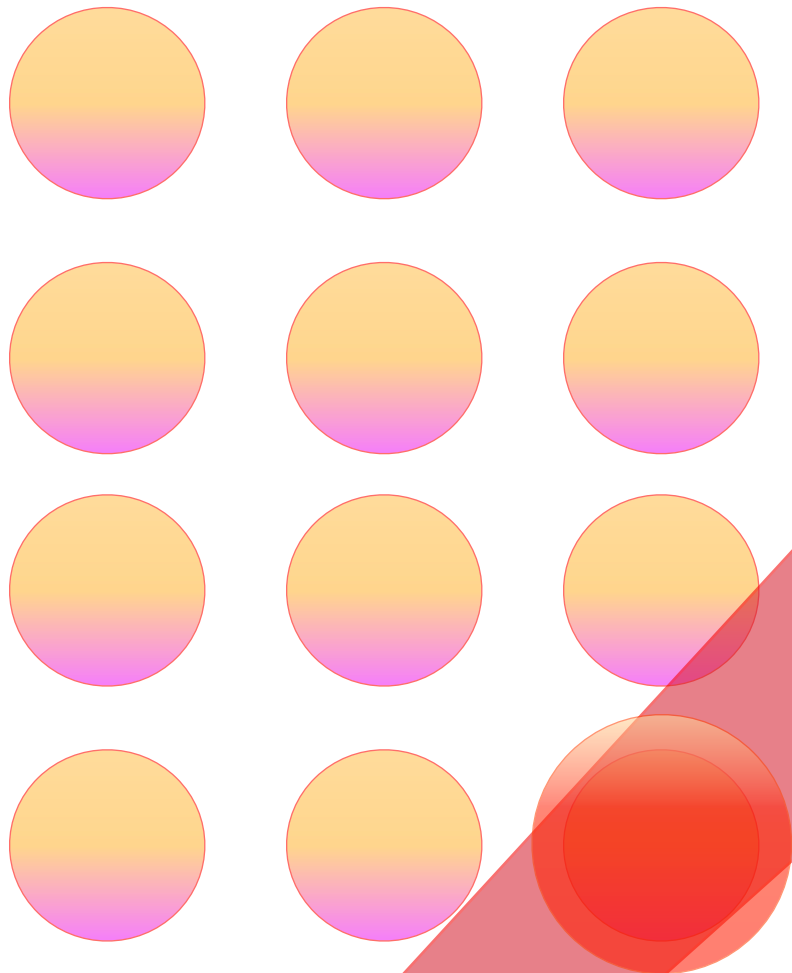




AOD

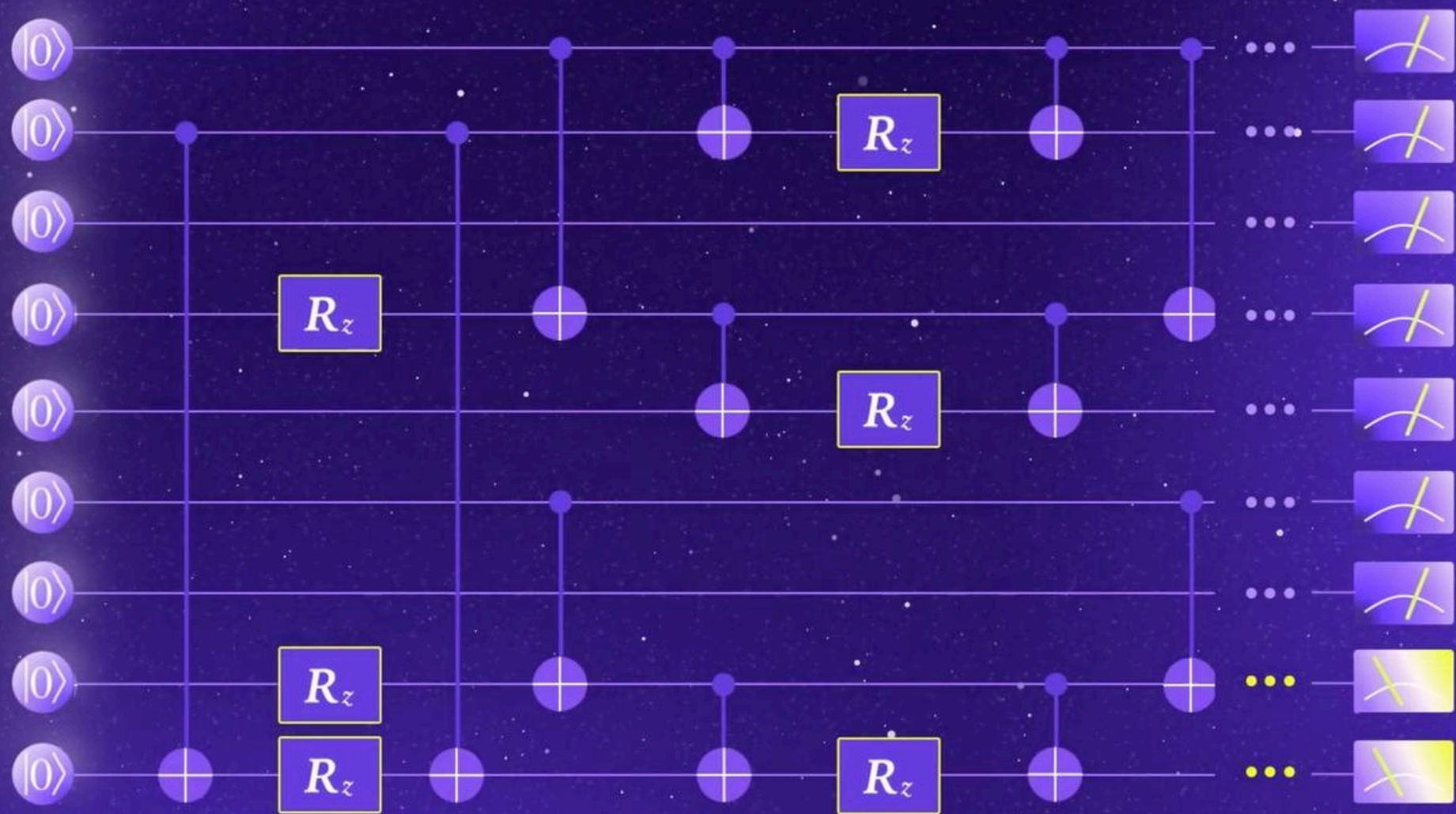
AOD

(Acousto- Optic
Deflector)

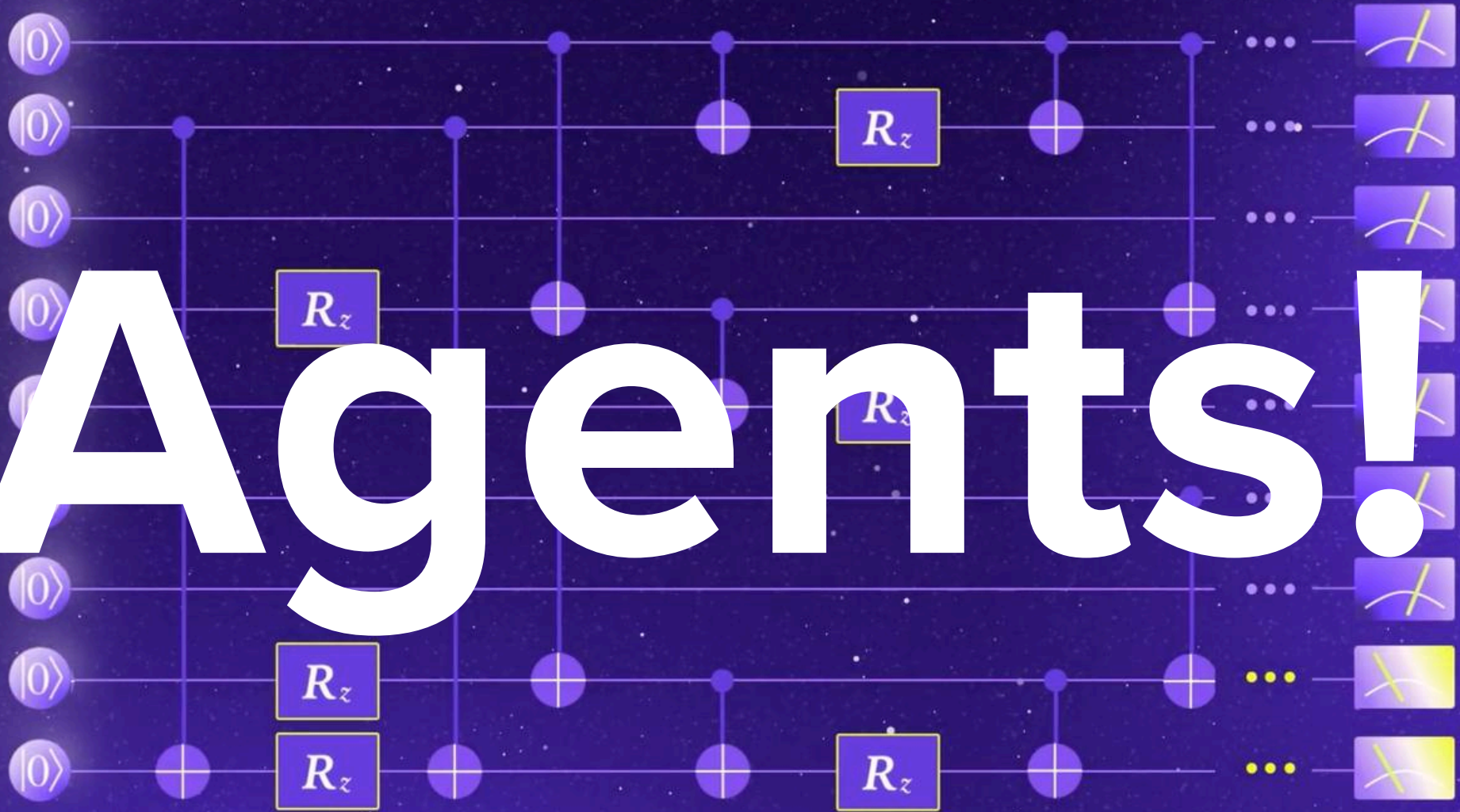


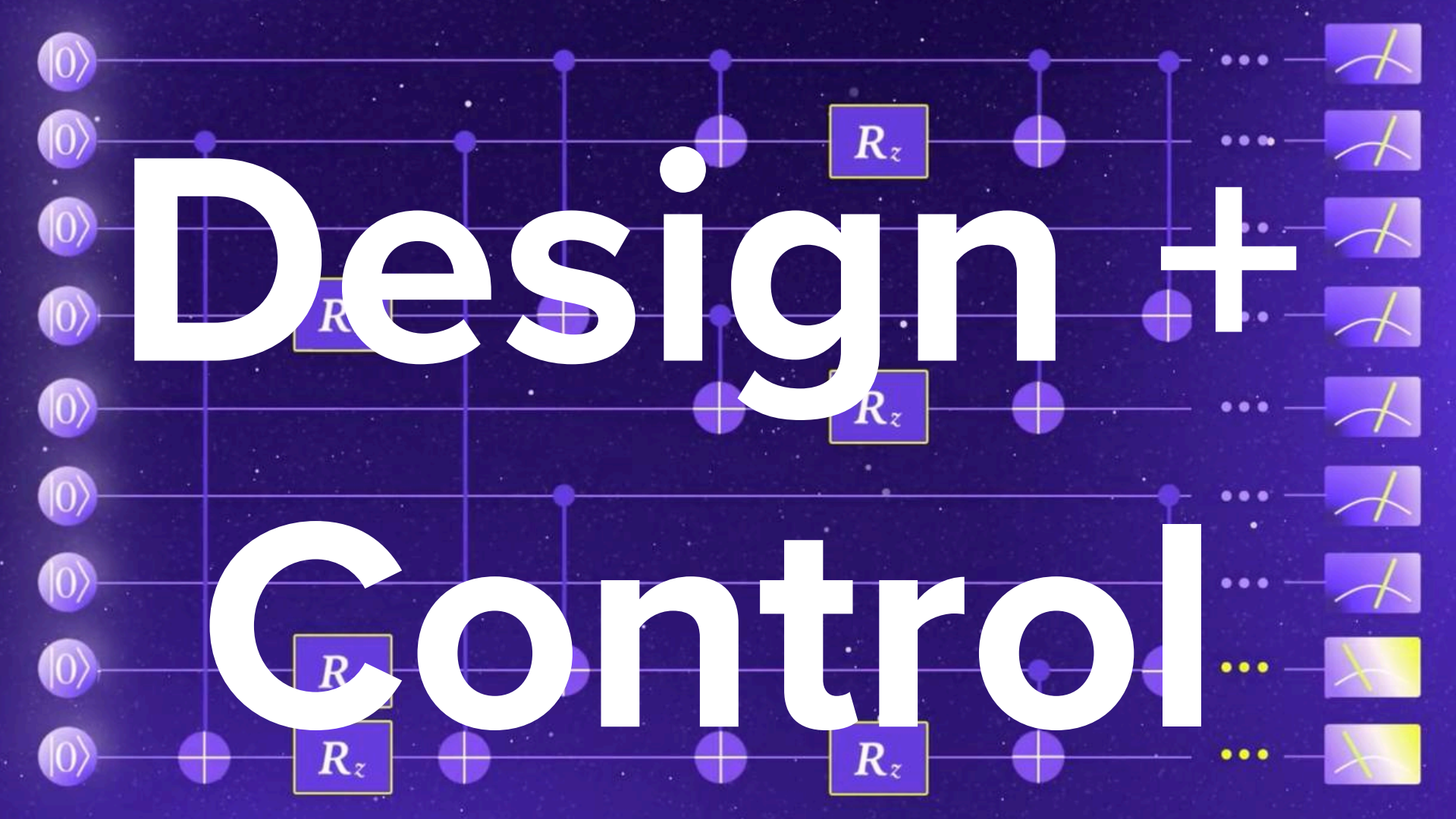
Gates & Circuits

Control the lasers



Agents!



A quantum circuit diagram on a dark blue background with a starry pattern. The circuit consists of multiple horizontal qubit lines, each starting with a $|0\rangle$ state in a purple circle. The lines are connected by vertical lines representing CNOT gates. Several gates are labeled R_z in yellow boxes. The circuit ends with measurement symbols on the right, represented by purple rectangles with a yellow diagonal line. The text "Design + Control" is overlaid in large white font.

Design + Control

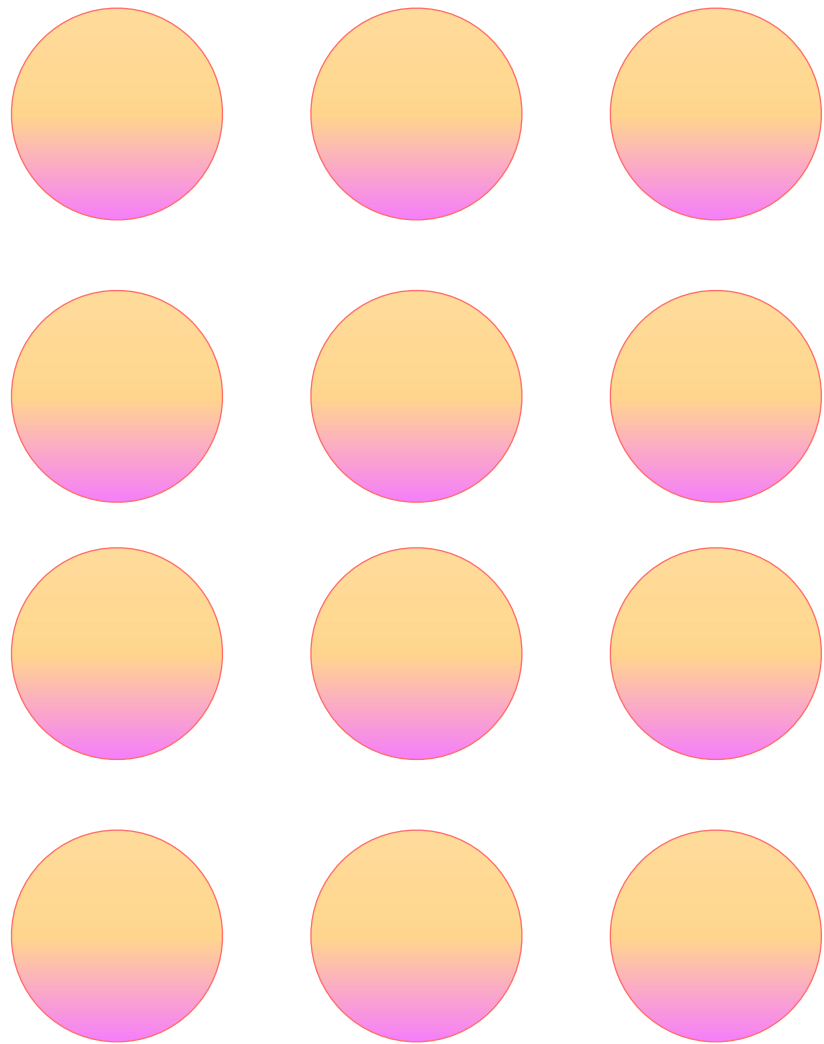
Agentic Control

- Code = Quantum Circuit
- Compile into voltages
- Drive the SLM/AOD

Readout

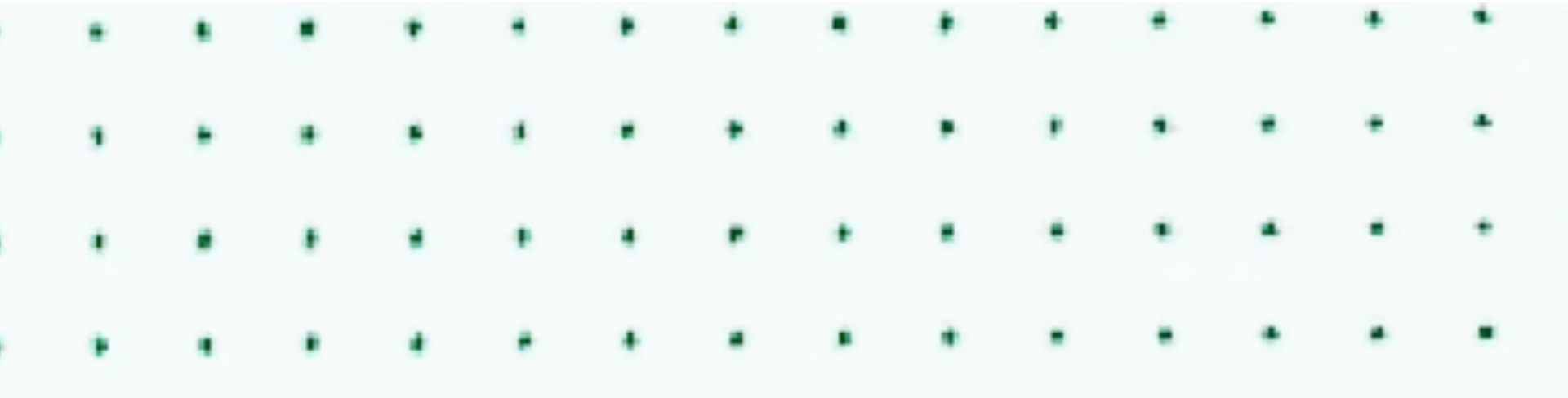
QPU output

Camera



The Qbits project 0 or 1

COMPUTER



VISION!

**ML is applied
on the
fluorescent
image**

Camera

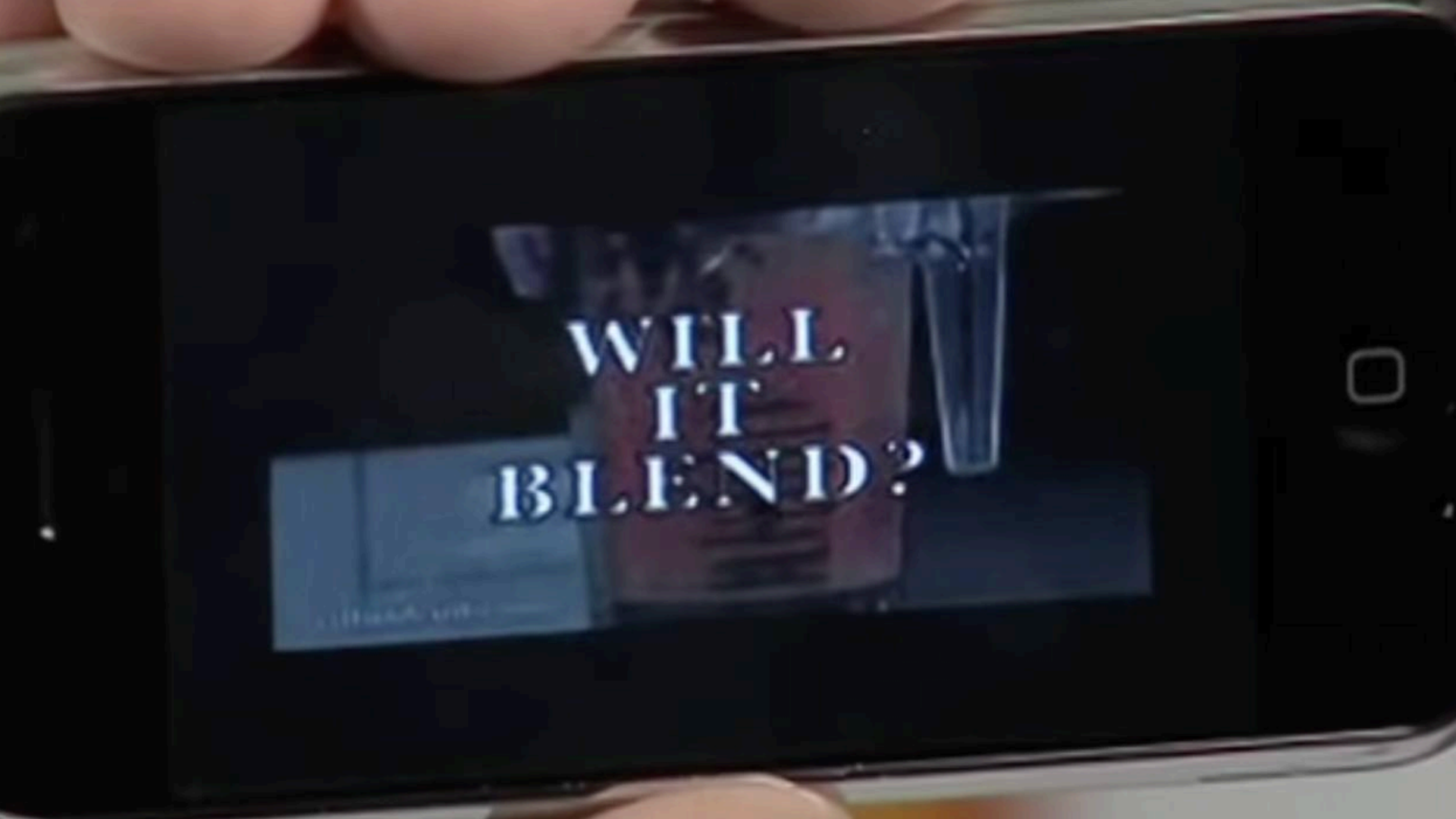
VL

MS

Camera

Agents!

Quantum +
AI?

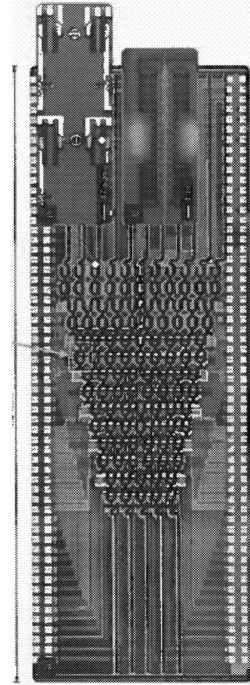
A close-up photograph of a smartphone held in someone's hand. The phone's screen is on, showing a video of a red blender jar. The text "WILL IT BLEND?" is superimposed in a white, serif font with a black outline. The phone is positioned over a wooden surface with several brown eggs. The phone's home button is visible on the right side of the device.

WILL
IT
BLEND?

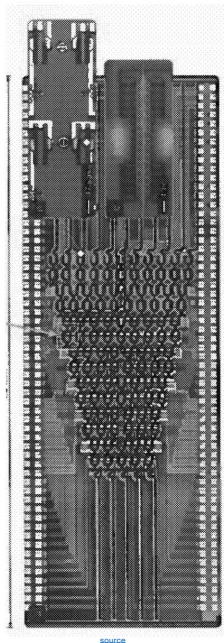
Yes!

ONNs

(Optical Neural
Networks)



Shen, Y., Harris, N., Skirlo, S. et al.
Deep learning with coherent
nanophotonic circuits. *Nature
Photon* **11**, 441–446 (2017).



Deep Learning on Photonic Integrated Circuits

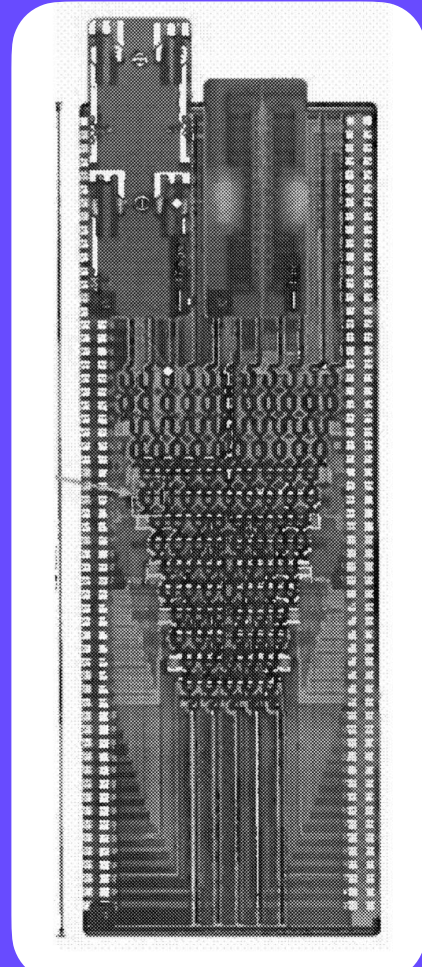
Dirk Englund's lab

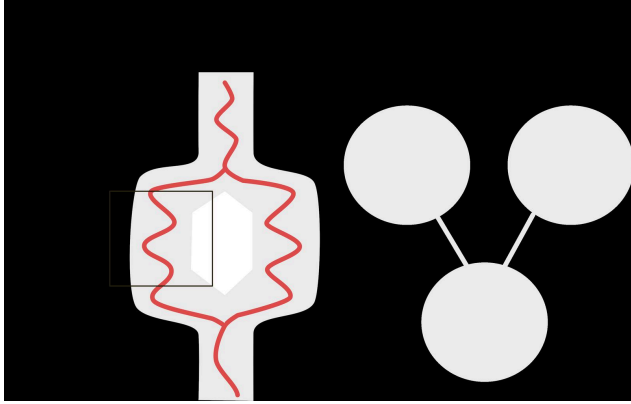


LIGHTMATTER



Optical Perceptron



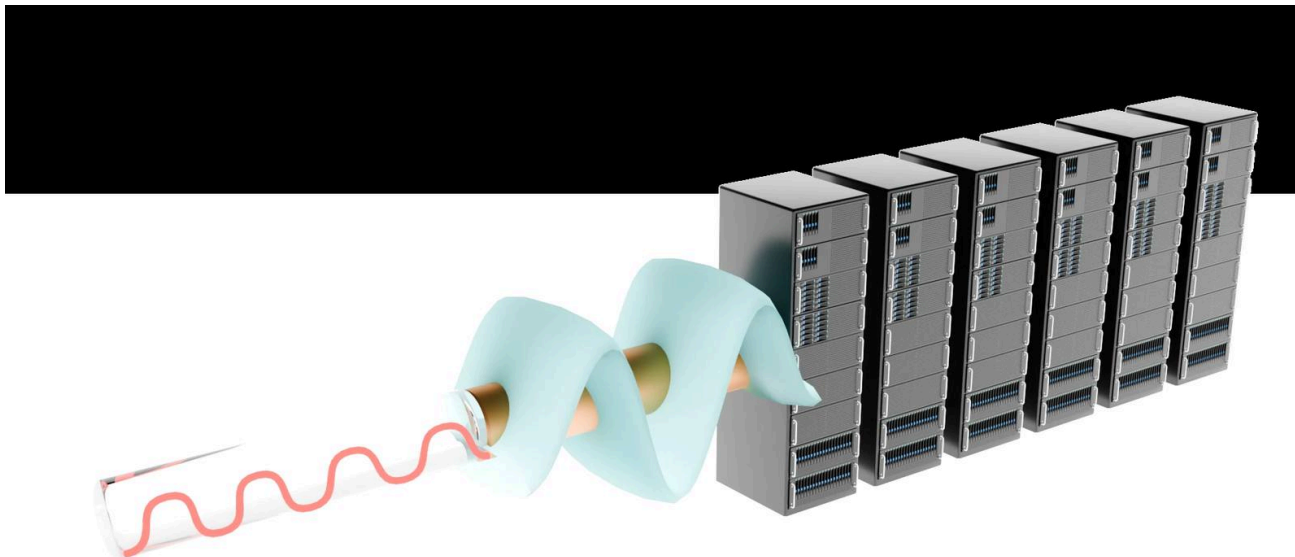


- Train parameters of phase shifters
- Reduce power consumption (once optical)
- Parallelize operations at light-speed

**Effective for
linear
operations,
room for
innovation
in non-linear
activation**

Problem

Photonic-Electronic Bottleneck

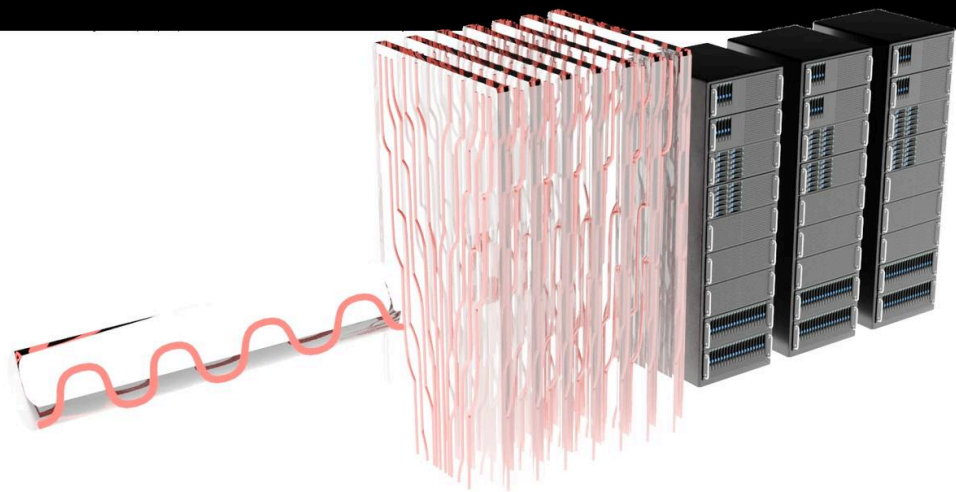
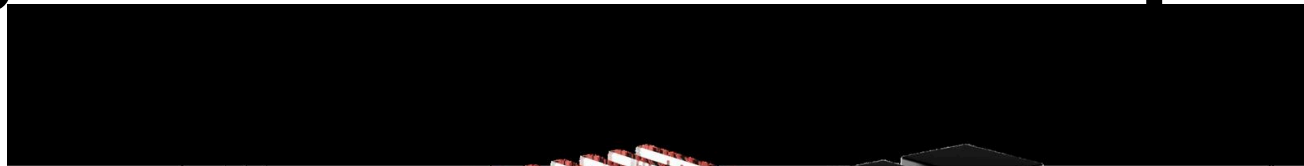


TelecomsData

ElectronicData

ElectronicCompute

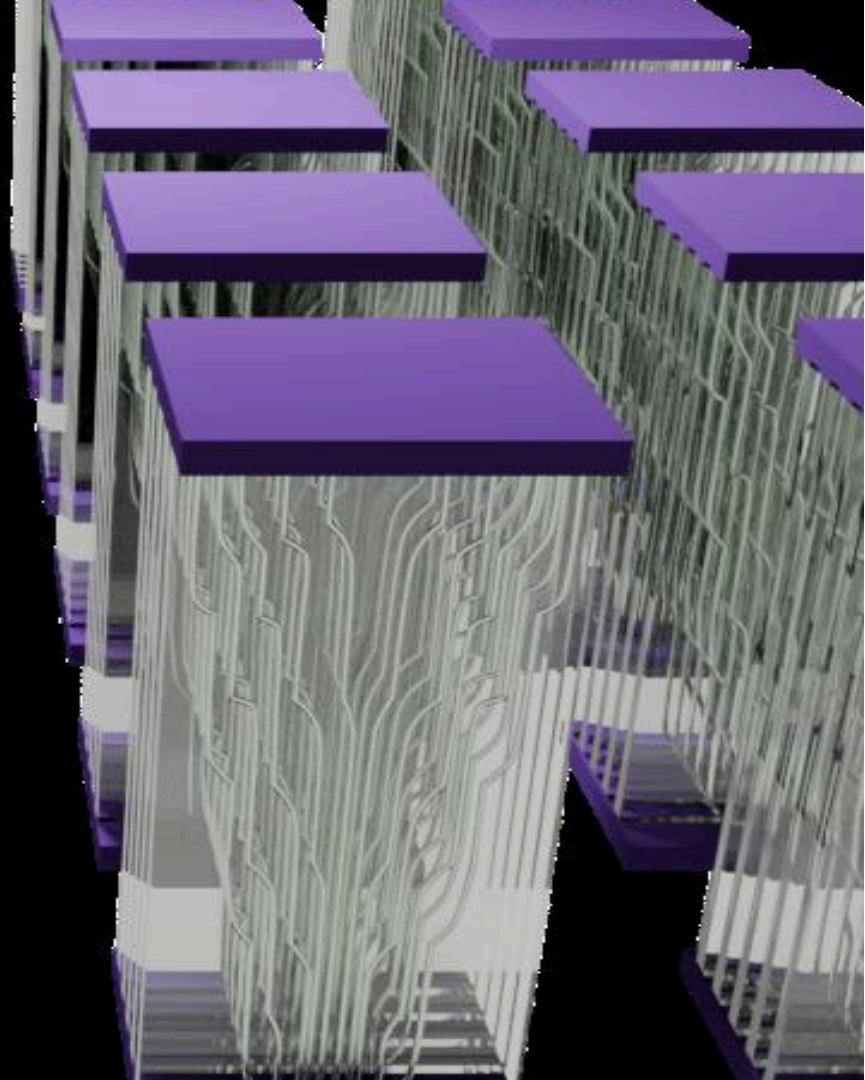
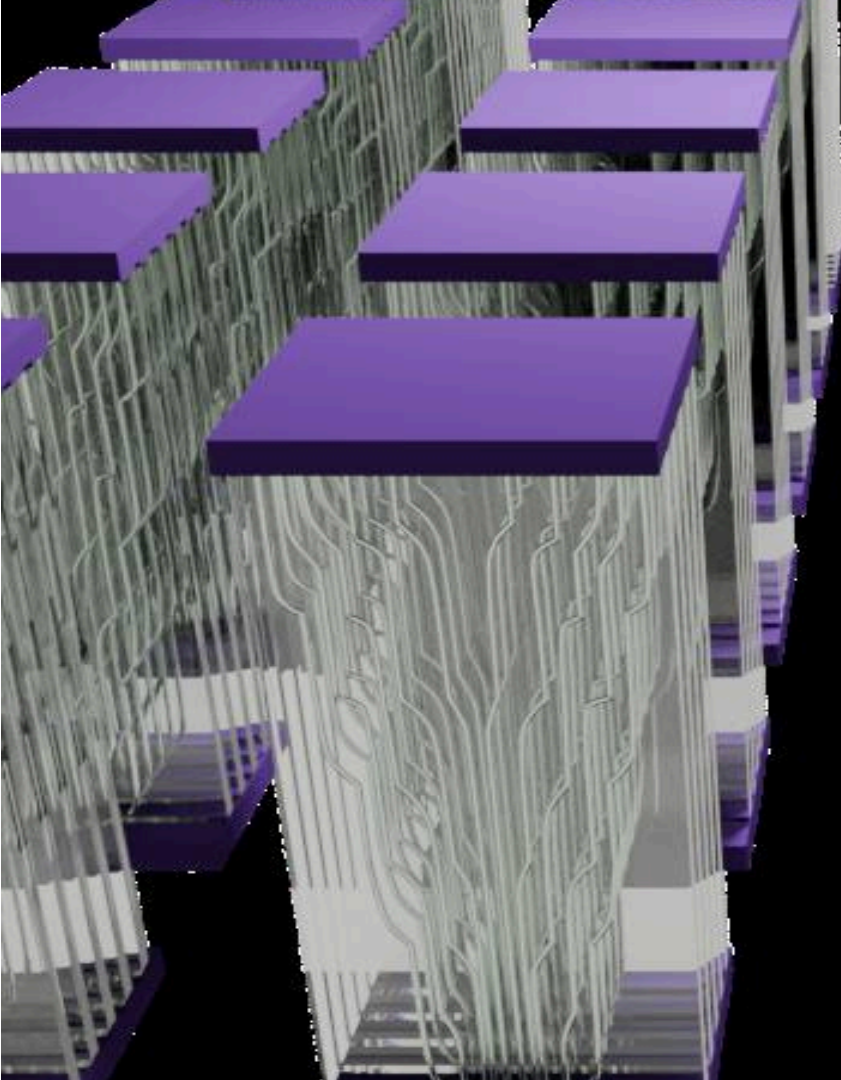
Hybrid-Electronic Compute

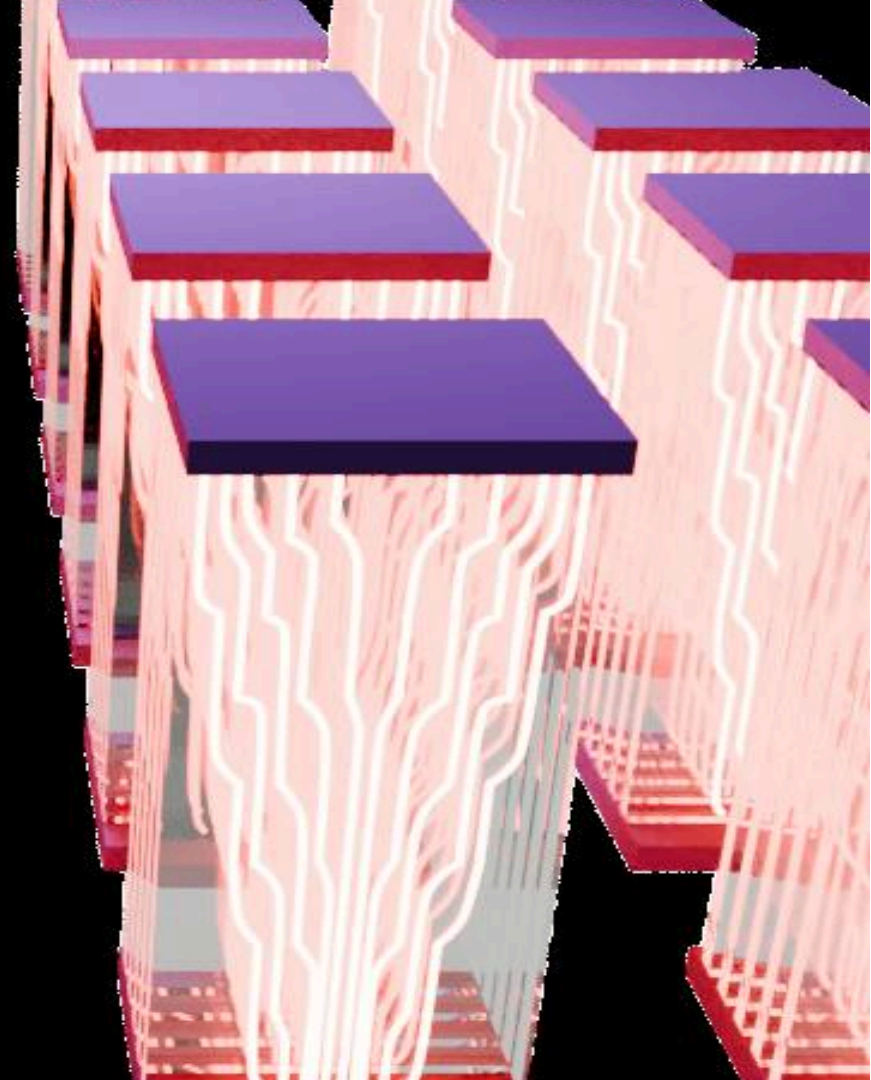
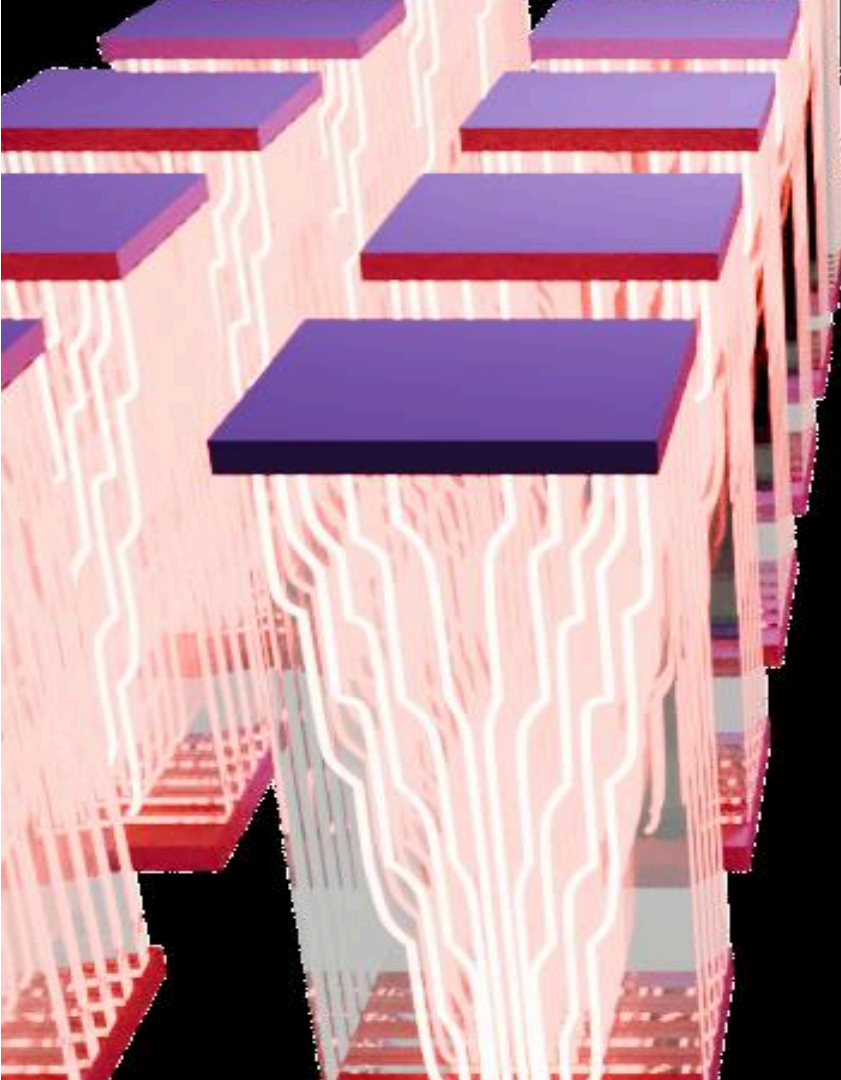


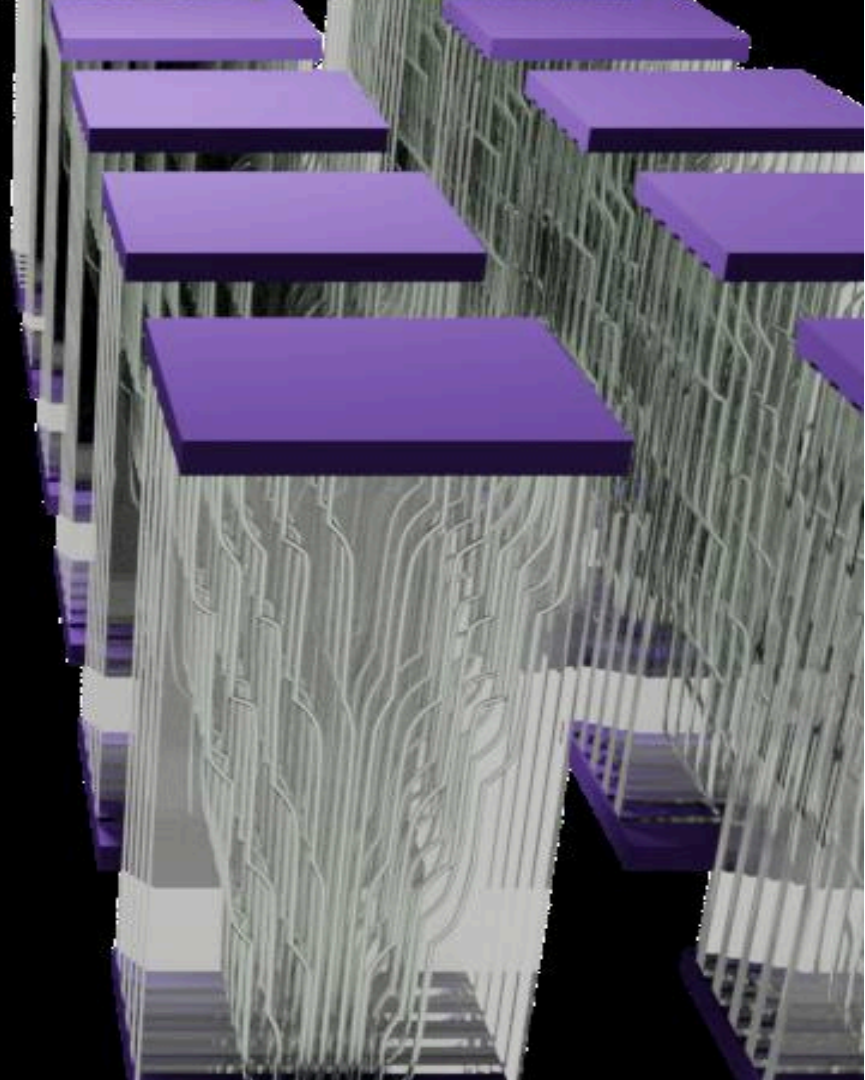
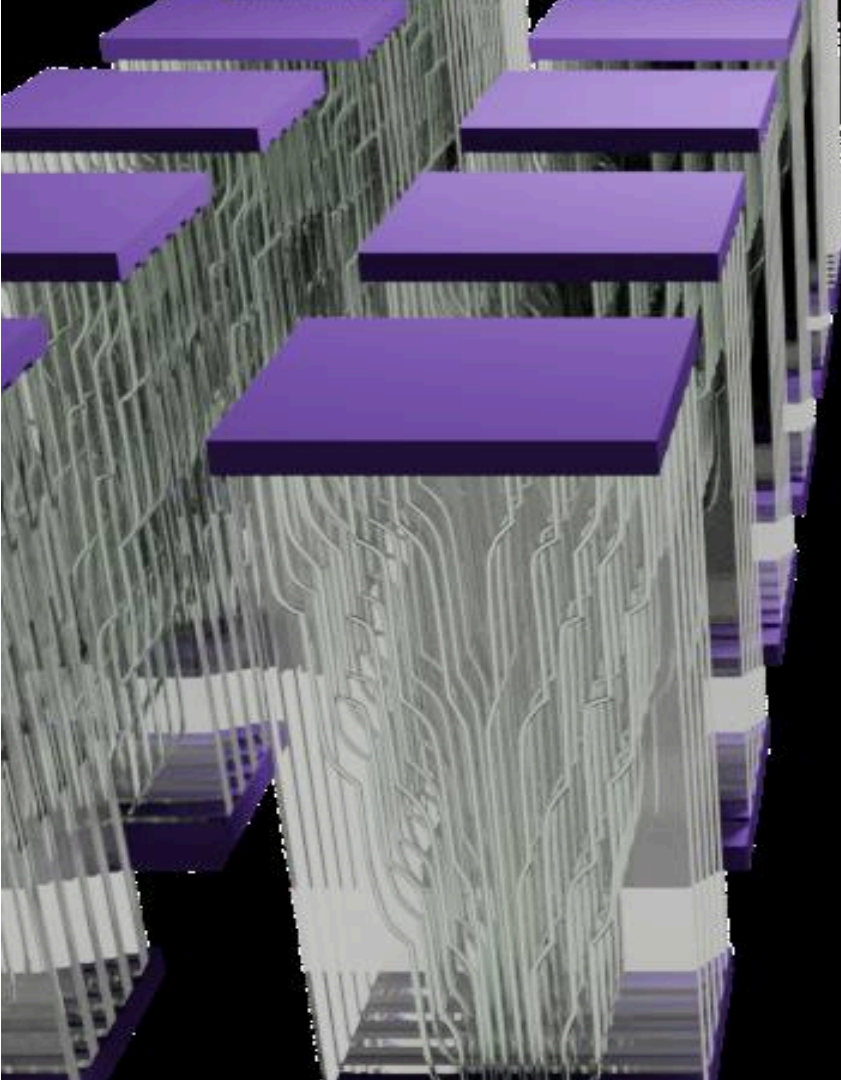
TelecomsData

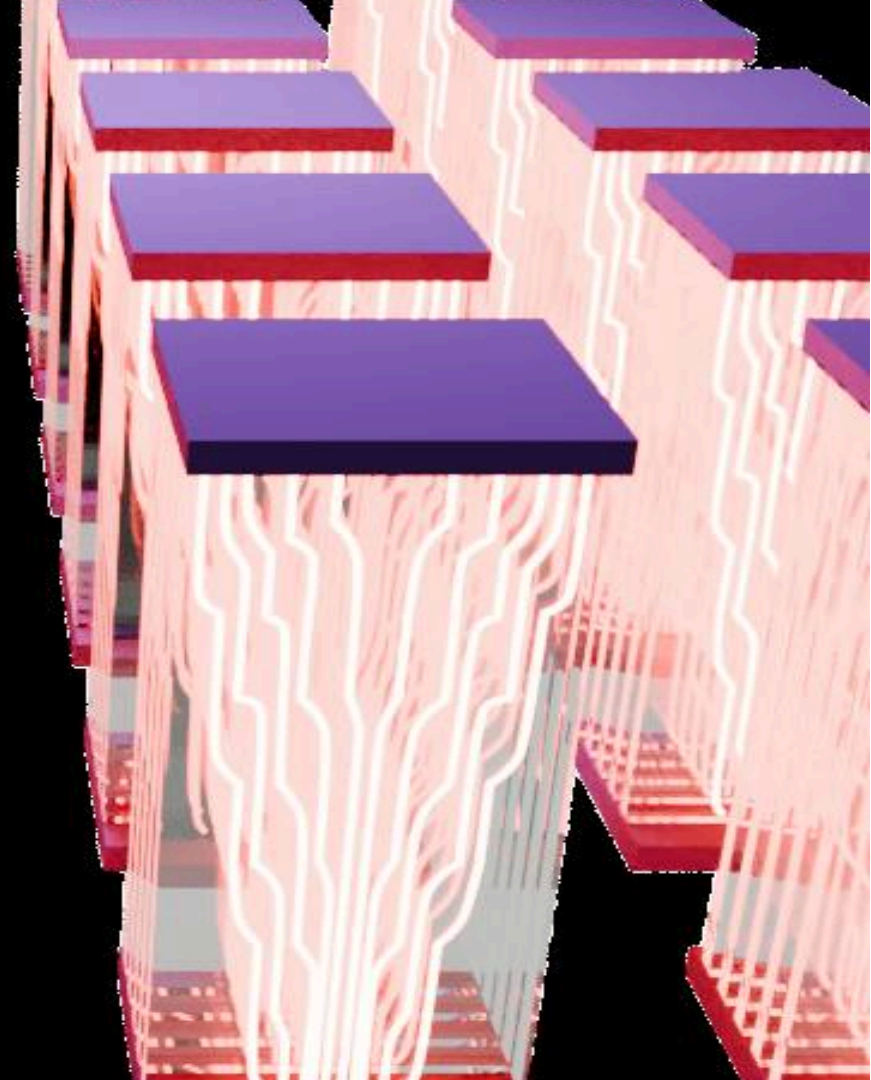
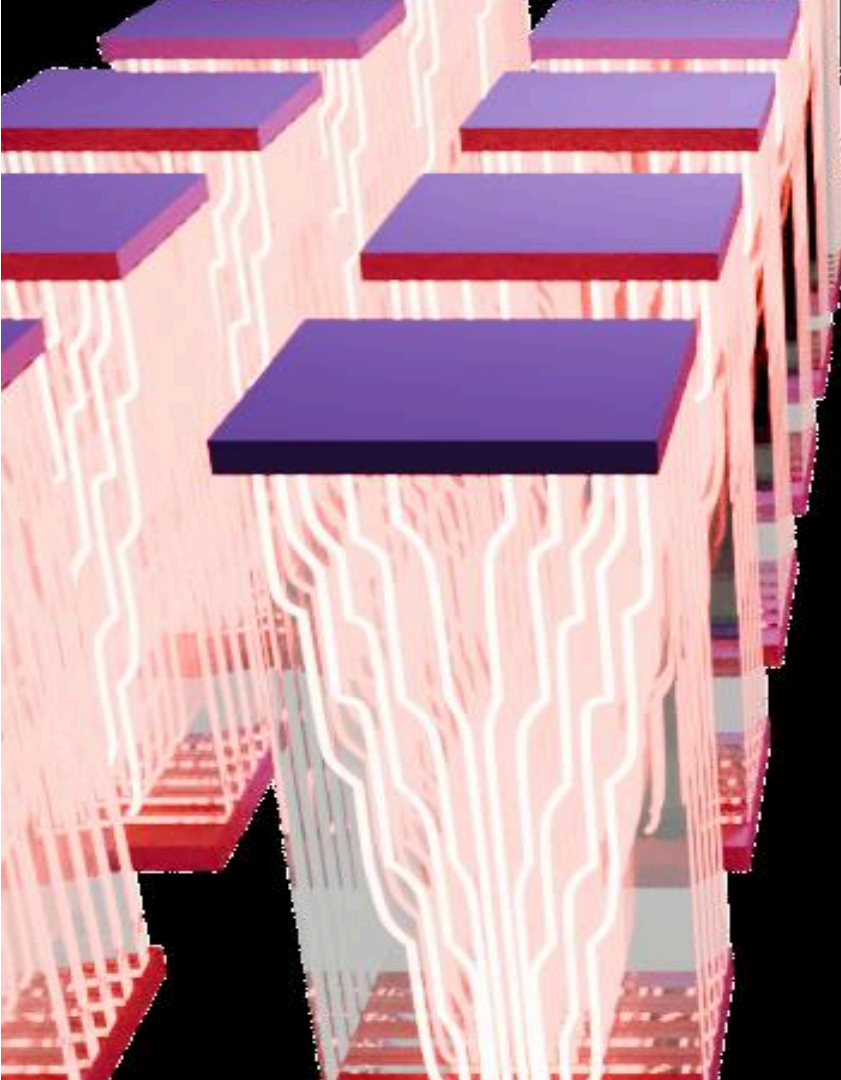
OpticalCompute

ElectronicCompute









**We don't have
to re-invent the
telecoms wheel**

**But! the
quantum-wheel
is literally being
invented**

Can ONNs
useful in
Quantum
computing?

Yes!

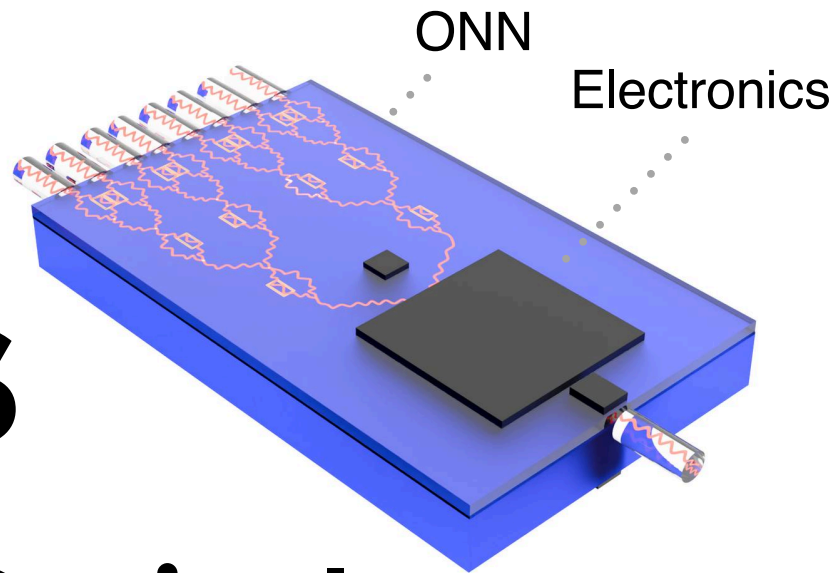
CPOs

(Co-Packaged Optics)

(Photonics + Electronics)

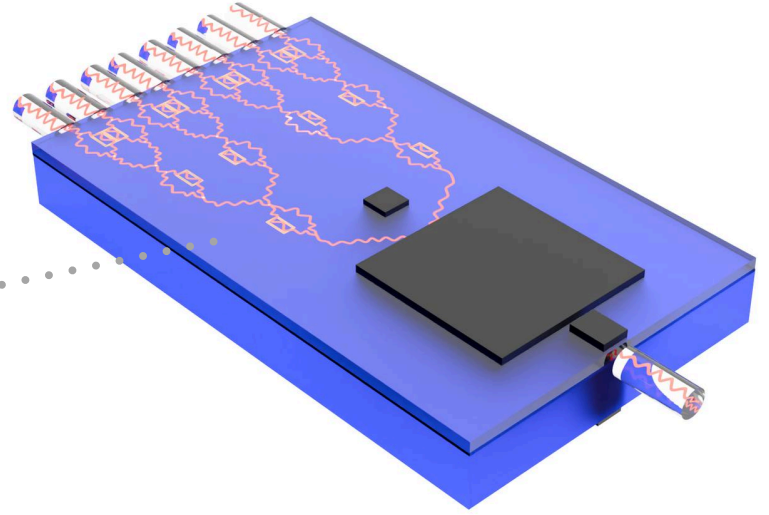
CPONNs

(Co-Packaged Optical
Neural Networks)



EO

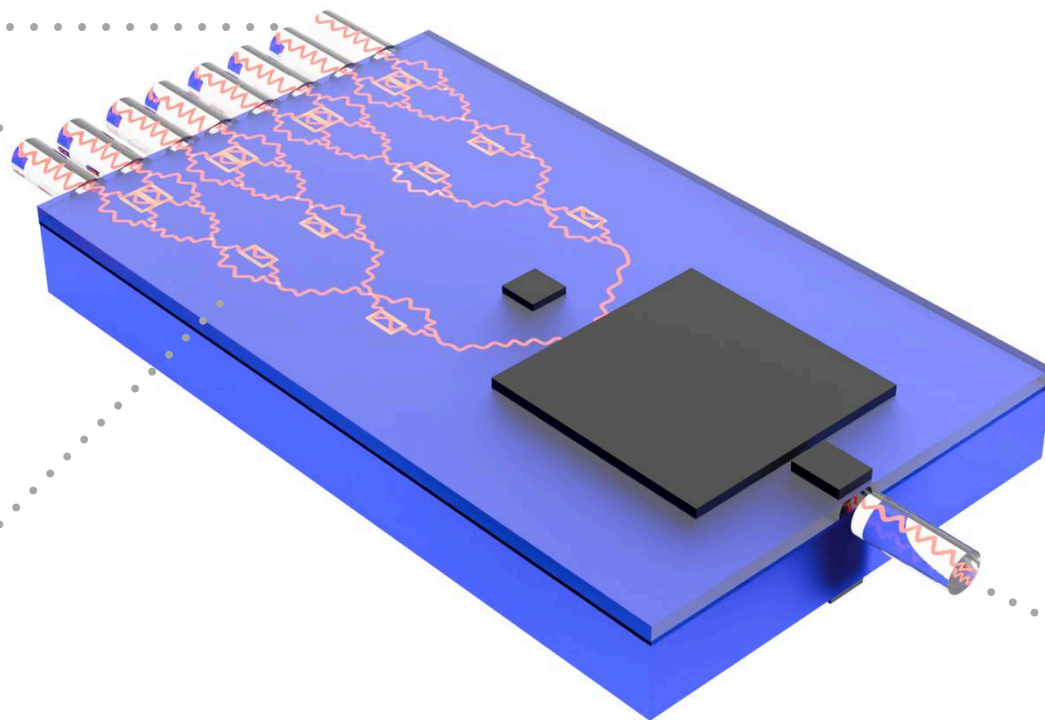
(Electro-Optic material)



CPONNs on Telecoms

Fiber Input

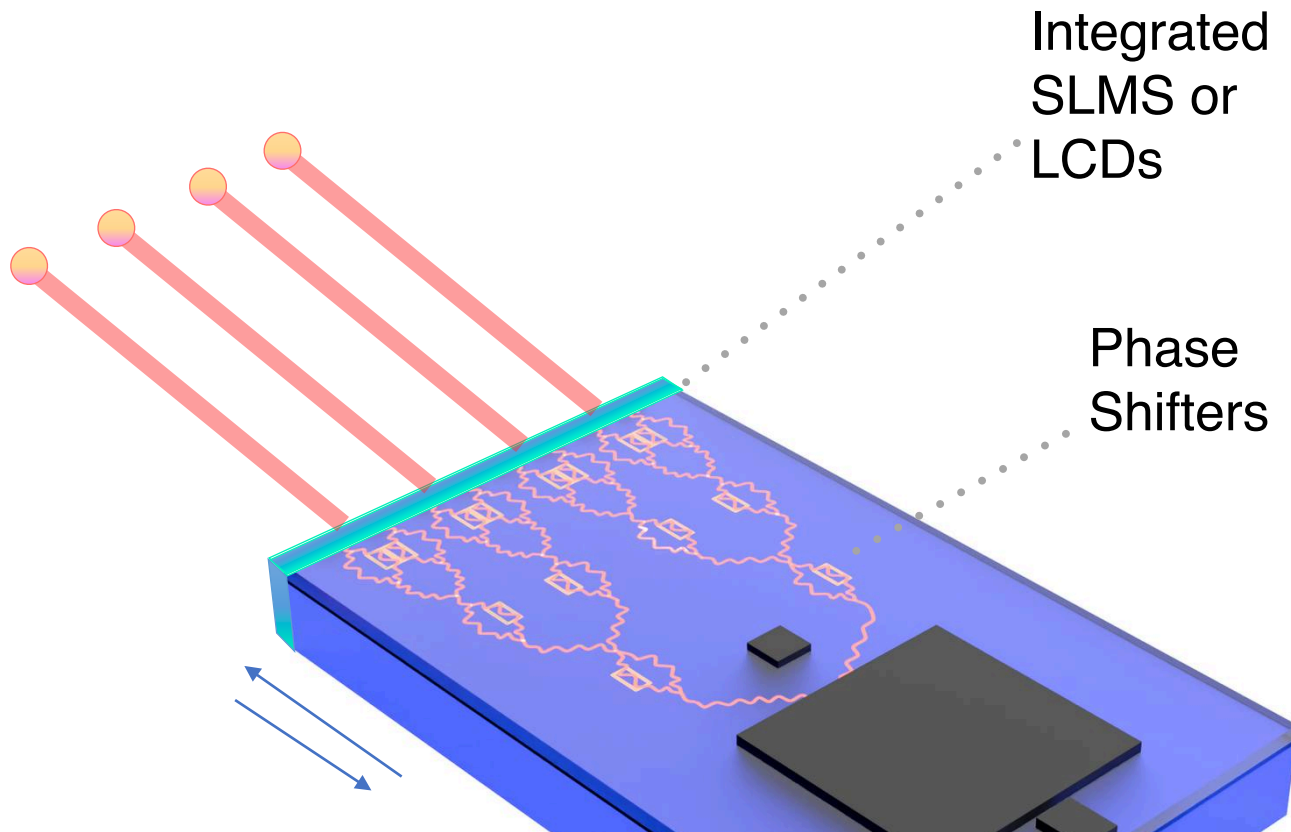
Trainable
Phase
Shifters



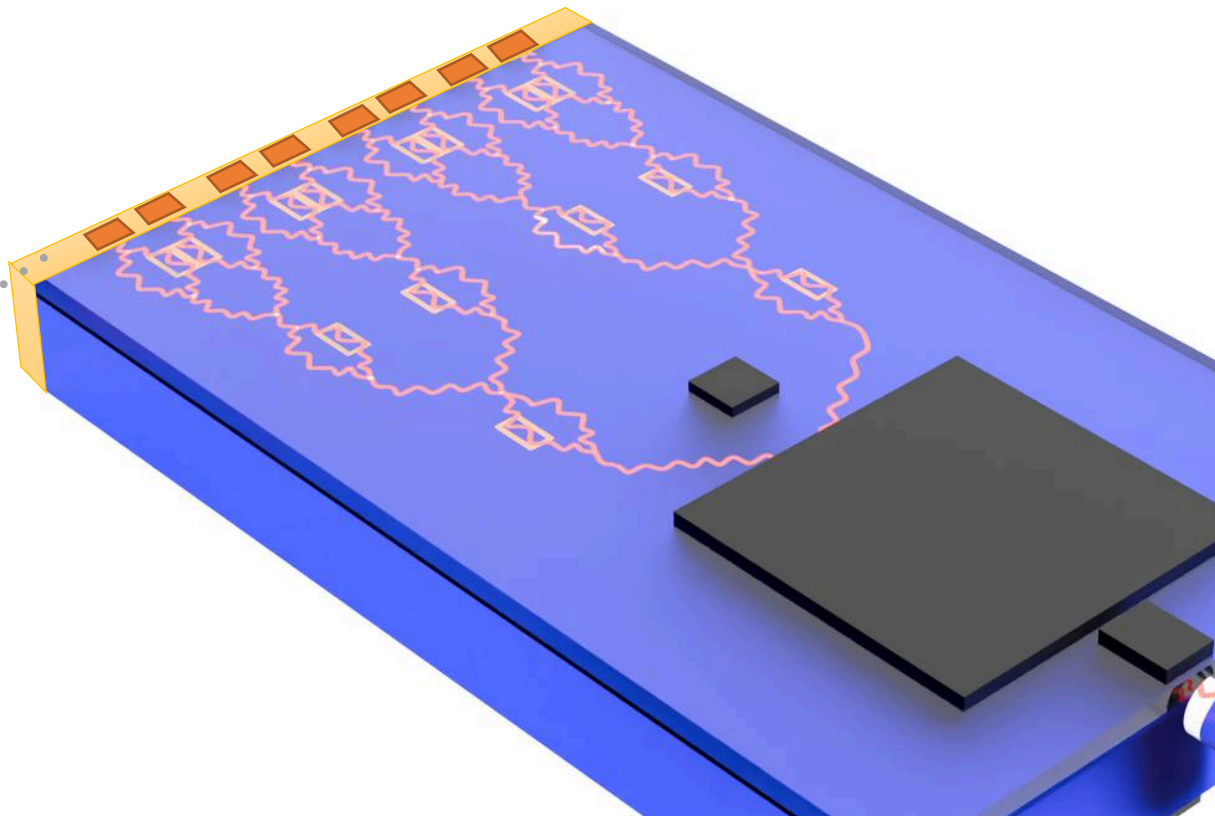
Fiber
Readout

ONNS on
Cold
Atoms?

Control CPONNs on Atoms



high-EO
effect
material



An integrated photonics platform for high-speed, ultrahigh-extinction, many-channel quantum control

Mengdi Zhao,^{1,*} Manuj Singh,^{1,*} Anshuman Singh,^{1,*} Henry Thoreen,^{1,*} Robert J. DeAngelo,¹ Daniel Dominguez,² Andrew Leenheer,² Frédéric Peyskens,¹ Alexander Lukin,¹ Dirk Englund,³ Matt Eichenfield,^{2,4} Nathan Gemelke,¹ and Noel H. Wan^{1,†}

¹*QuEra Computing Inc, Boston, MA, 02135, USA*

²*Sandia National Laboratories, Albuquerque, NM, 87185, USA*

³*Massachusetts Institute of Technology, Cambridge, MA, 02139, USA*

⁴*Wyant College of Optical Sciences, University of Arizona, Tucson, AZ, 85721, USA*

(Dated: August 14, 2025)

High-fidelity control of the thousands to millions of programmable qubits needed for utility-scale quantum computers presents a formidable challenge for control systems. In leading atomic systems, control is optical: UV–NIR beams must be fanned out over numerous spatial channels and modulated to implement gates. While photonic integrated circuits (PICs) offer a potentially scalable solution, they also need to simultaneously feature high-speed and high-extinction modulation, strong inter-channel isolation, and broad wavelength compatibility. Here, we introduce and experimentally validate a foundry-fabricated PIC platform that overcomes these limitations. Designed for Rubidium-87 neutral atom quantum computers, our 8-channel PICs, fabricated on a 200-mm wafer process, demonstrate an advanced combination of performance metrics. At the 795 nm single-qubit gate wavelength, we achieve a mean extinction ratio (ER) of 71.4 ± 1.1 dB, nearest-neighbor on-chip crosstalk of -68.0 ± 1.0 dB, and -50.8 ± 0.2 dB after parallel beam delivery in free-space. This high-performance operation extends to the 420 nm and 1013 nm wavelengths for two-qubit Rydberg gates, showing ERs of 42.4 dB (detector-limited) and 61.5 dB, respectively. The devices exhibit 10-90% rise times of 26 ± 7 ns, achieve dynamic switching to -60 dB levels within microsecond timescales, and show pulse stability errors at the 10^{-3} level. This work establishes a scalable platform for developing advanced large-scale optical control required in fault-tolerant quantum computers and other precision technologies.



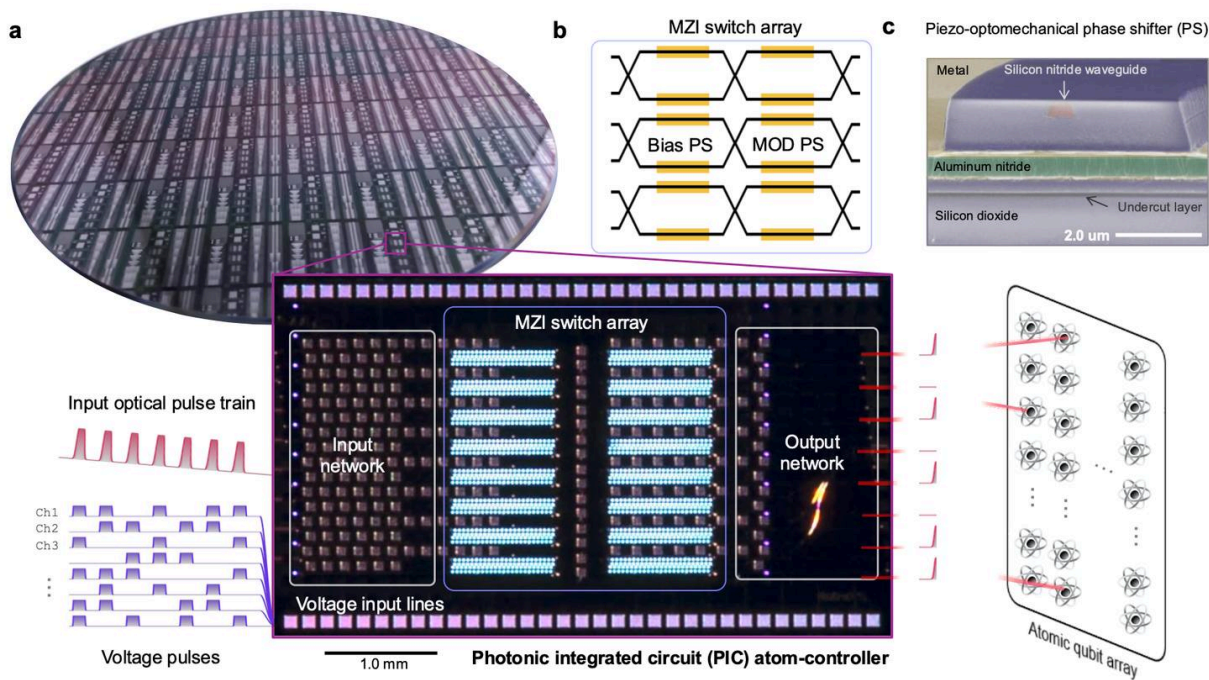
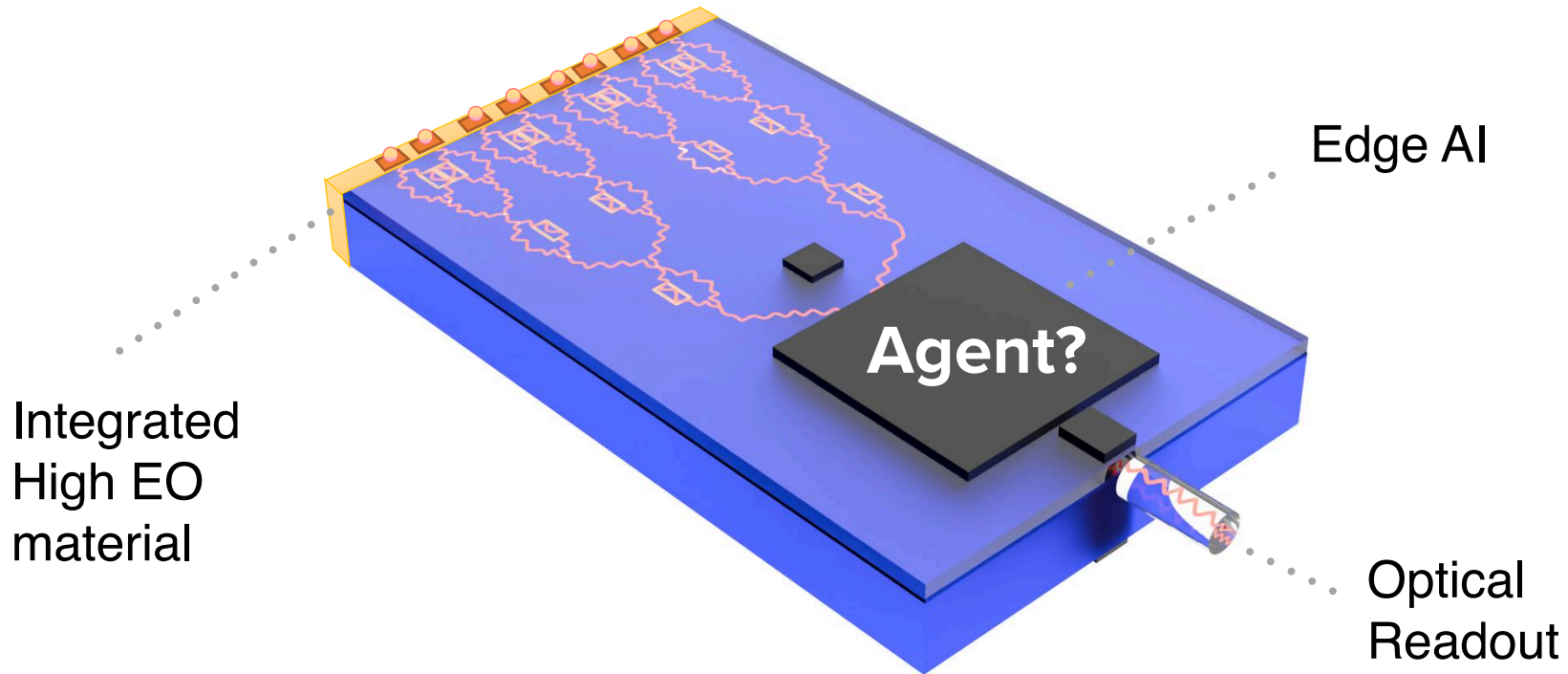


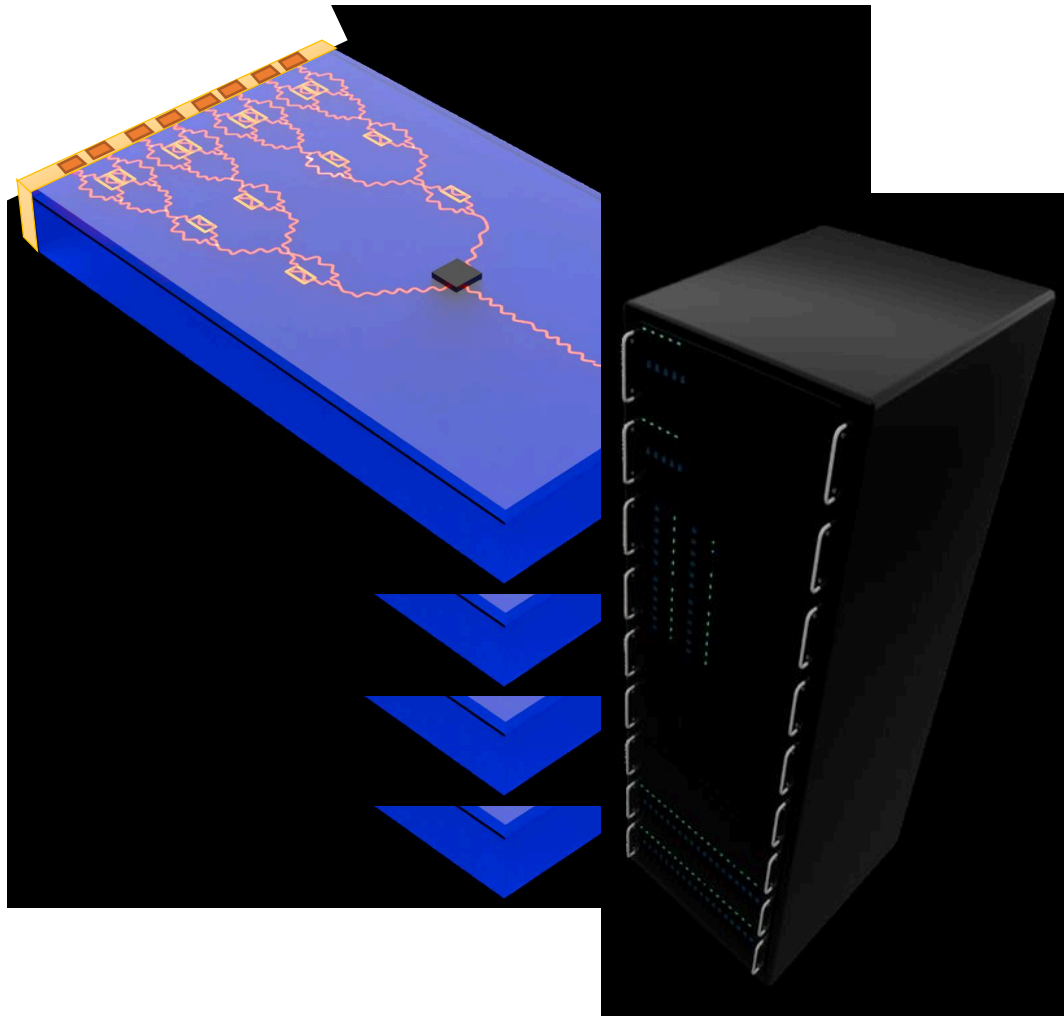
Figure 1: **A scalable integrated photonics-based architecture for high-fidelity atomic quantum control.** **a**, A fully processed 200 mm wafer for precise optical addressing of atomic qubit arrays. The central optical micrograph shows a single, multichannel atom-controller in a photonic integrated circuit (PIC) that transduces electronic waveforms into precise optical pulse trains for parallel, individual qubit addressing. **b**, As the core functional unit, each programmable MZI switch incorporates phase shifters (PS) for both static bias tuning and high-speed modulation (MOD). **c**, False color scanning electron micrograph image of the underlying piezo-optomechanical technology, which consists of an integrated aluminum nitride (AlN) actuator for high-bandwidth phase modulation in the mechanically coupled silicon nitride (SiN) waveguide.

Edge AI,
on chip?

CPONNs on atom Qbits



**Agentic
orchestration?**



**Agentic
CPONN-
controlled
Quantum
Computer?**

AI!

Where?

- Circuit design
- On edge, for control
- Agent Orchestrator (Control/Readout)

Dr. Kevin Mato



Hardware Matters!

Calibration is a full stack engineering problem!

Total round-trip (measurement to pulse update):
1-3 μ s
(achievable on FPGA)

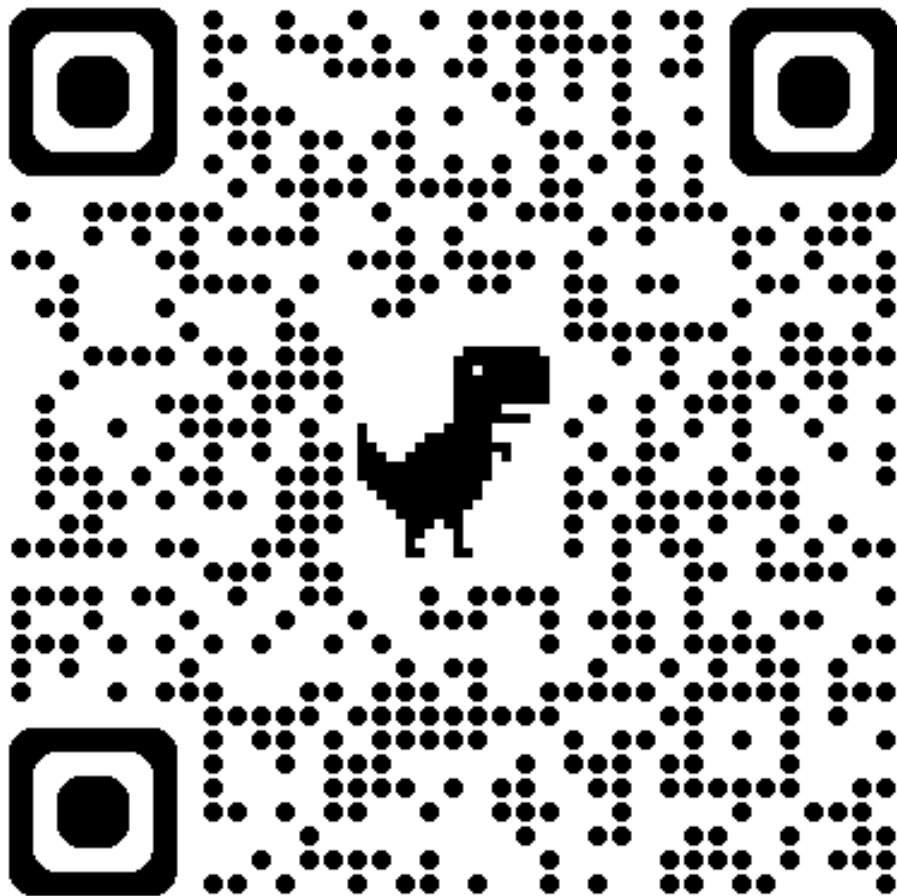
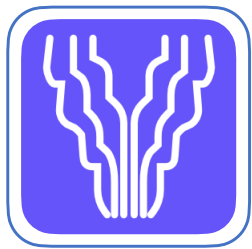
AI?
Self tuning?

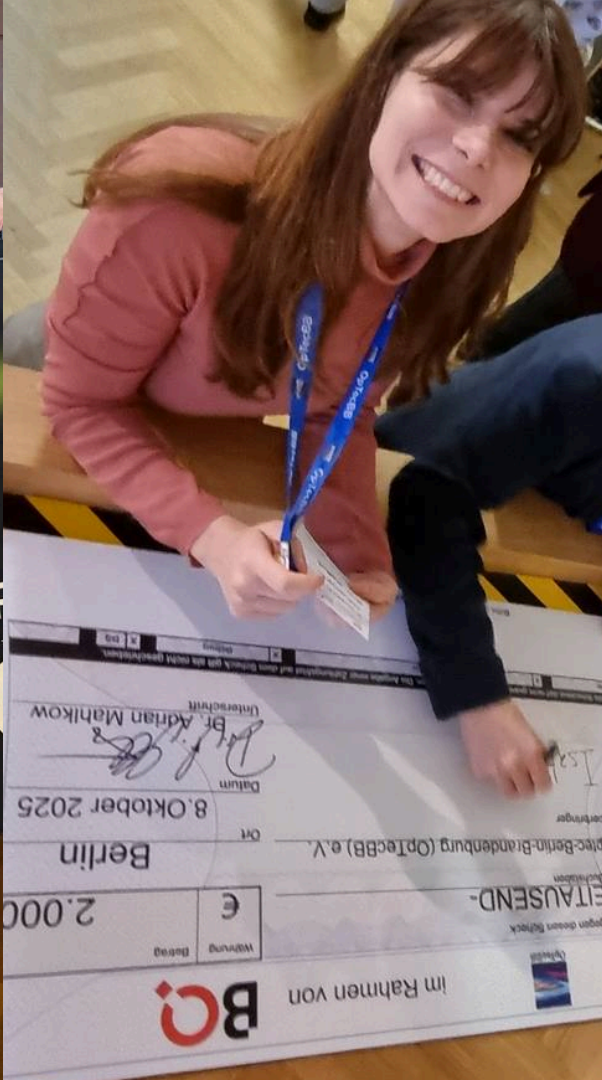
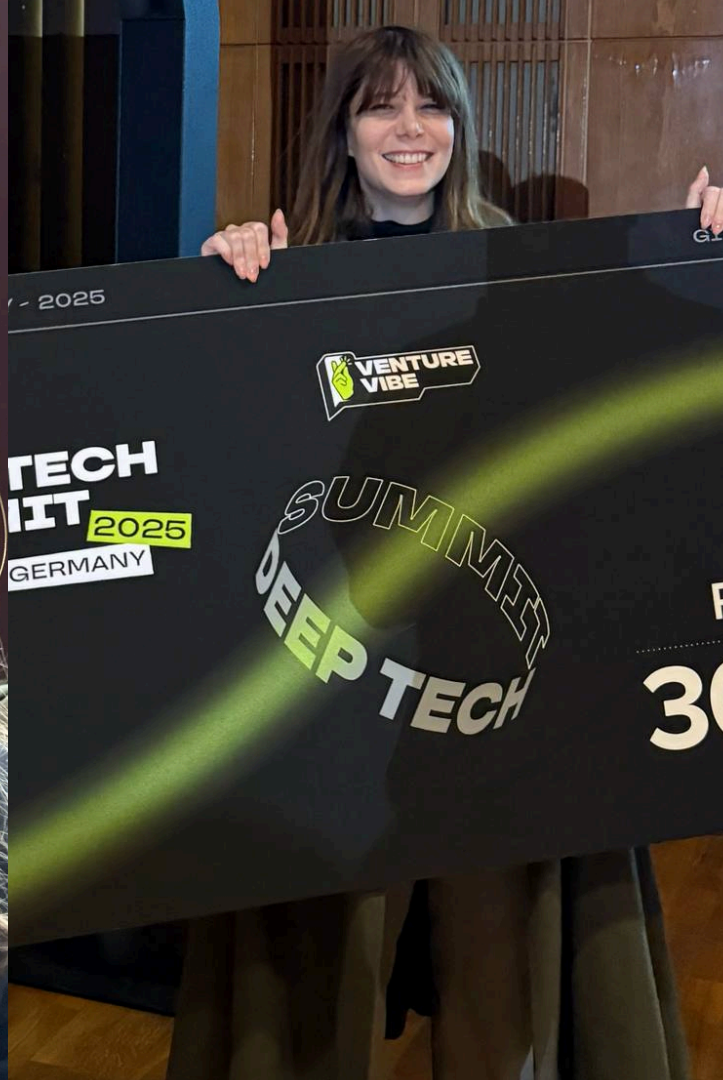
YouTube

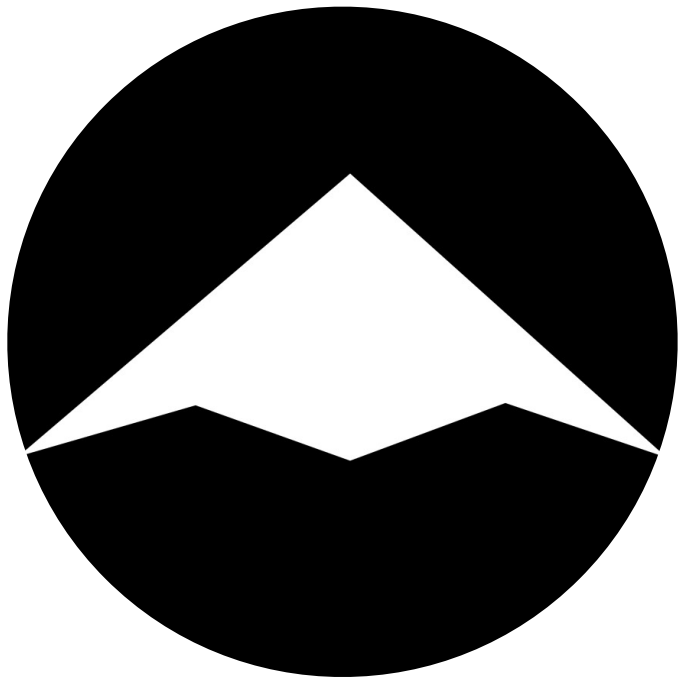
Quantum Café Berlin March 28th 2026

laser neuron™
5 subscribers

3 Share Ask Save Download



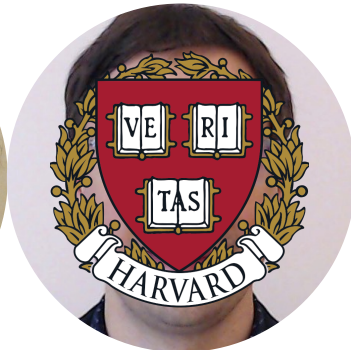




Founders



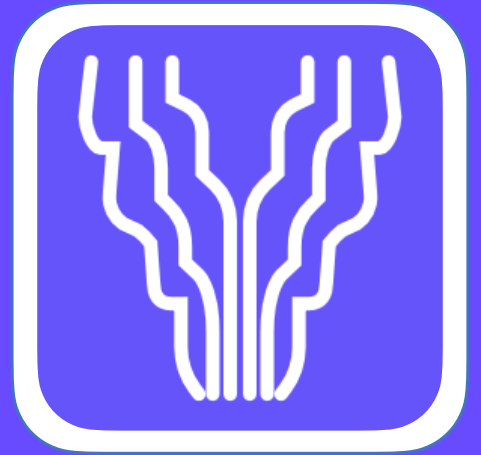
Stors



Science
belongs to
everyone

Science is
useful when it
meets a market

Connecting Scientists with VC



laser neuron™

Panels
Hacks
Cafés
Workshops

Berlin



Quantum Café
Berlin

Saturday March 28th, 2026 @ CIC Berlin 3pm-6pm

OpTecBB  BQ   leap  laser neuron™ 

This poster features seven circular portraits of speakers arranged in two rows. The top row has three portraits, and the bottom row has four. The background is a solid blue color.

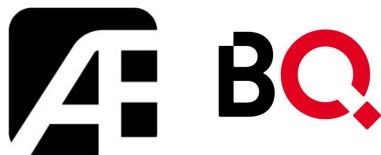


Quantum
Café
Berlin

Saturday June 6th, 2026 @ CIC Berlin 3pm-6pm

 laser neuron™  leap  OpTecBB  BQ 

This poster features five circular portraits of speakers. Four are in the top row and one is in the bottom right. The background is a solid blue color.



Berlin

**Isabella
Douzoglou**



**Dr. Ingmari
Tietje**



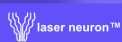
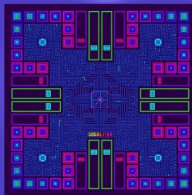
**Dr. Jana
Lehner**



Boston

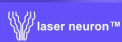
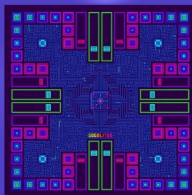
Quantum Un-Conference

Saturday February 21st, 2026
@ CIC Cambridge 2-7pm



Sundai Quantum Hack

Sunday February 22nd, 2026
@ MIT IHQ 10am-10pm



BOS TECH WEEK



we're
hosting
HACKING



MIT HOUSE



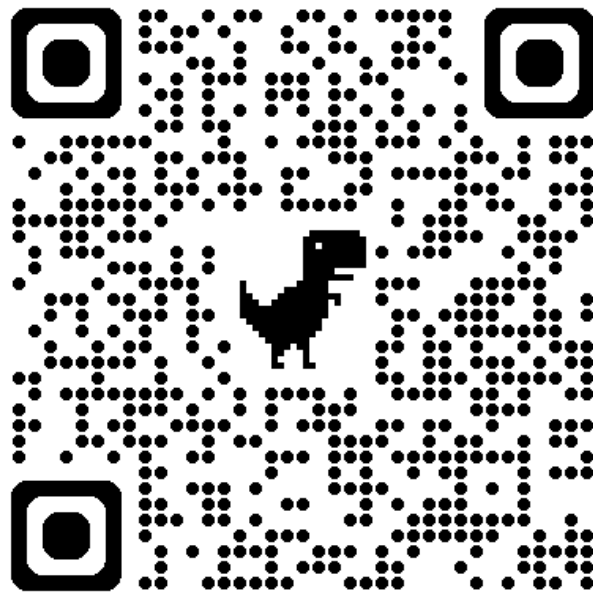
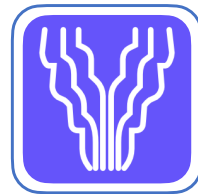
Weights
& Biases



CoreWeave

Boston

Isabella
Douzoglou



Contact me if you're interested
in getting involved!
isabella@laserneuron.ai

There will be more!



laser neuron™ Berlin

137 Subscribers



laser neuron™ Boston

99 Subscribers

Thank you!
Questions?
:D

isabella@laserneuron.ai